GNN for PhC

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Repository: https://github.com/qinlin-kip/GNN-PhC

Generative Neural Networks for the Sciences

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Abstract

Traditionally, discovering high-performing photonic devices relies on computationally expensive physics simulations. Furthermore, the "inverse problem" of mapping a target optical spectrum back to a physical geometry is notoriously ill-posed; because multiple geometries can produce the same spectrum, standard deterministic models often fail by outputting blurred, physically unbuildable averages. We addressed this through a multi-part data-driven pipeline using fabrication-constrained datasets. First, we solved the forward problem by training robust Convolutional Neural Networks (CNNs) and Multi-Layer Perceptrons (MLPs) to near-instantaneously predict bandgaps, reflection, transmission, and scattering losses, effectively replacing slow physics simulators. In the waveguide-focused part of the project, we trained image-based models for predicting reflection, transmission, and loss-related optical responses from photonic-crystal geometries, and also explored inverse models that generate candidate structures from target spectral behavior. Second, we tackled the inverse design challenge by comparing three distinct probabilistic and constrained approaches: a conditional Variational Autoencoder (cVAE) for stochastic candidate generation, a Geometry Autoencoder combined with an MLP surrogate for gradient-based Latent-Space Optimization, and physics-constrained Denoising Diffusion Probabilistic Models (DDPMs). These methods successfully generated valid, high-performing structural blueprints. Crucially, we identified a "Metric Misalignment Paradox": optimizing models purely for mathematical accuracy (e.g., Mean Squared Error) forces them to reproduce jagged, unmanufacturable simulation artifacts, whereas our constrained latent and generative models naturally yield smooth, cleanroom-ready designs. Currently, our pipeline successfully proposes candidate geometries evaluated via neural network surrogates. Validating these AI-generated structures through full-wave electromagnetic simulations and experimental cleanroom fabrication remains the primary focus for future work.

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1 Introduction

The Bottleneck in Traditional Photonics
The design of photonic crystals—materials that dictate the flow of light—has historically relied on rigorous, theory-driven computational solvers. While these physics-based simulations provide extremely accurate predictions of how light interacts with a given material, they are computationally expensive and time-consuming. This creates a severe bottleneck for innovation: attempting to discover an optimal, high-performing design through a traditional trial-and-error search of complex geometries is often prohibitively slow [7, 5].

Building a Realistic Data Foundation
To bypass these computational limits, our project utilizes a data-driven machine learning approach. The essential first step is generating a robust training dataset comprising hundreds of realistically fabricable structural geometries. We then use conventional, high-fidelity physics solvers to calculate the exact optical characteristics (such as band structures, spectrum, scattering loss) for each individual design. This establishes a comprehensive, paired library of physical shapes and their corresponding optical behaviors to serve as the ground truth for training our networks [2, 5].

The Forward Problem: High-Speed Predictive Networks
Using this dataset, we first train a predictive neural network to solve the "forward problem." This network acts as a highly efficient stand-in for the traditional simulator, learning to map a structure's physical geometry directly to its resulting optical properties. Once the network understands these underlying geometric relationships, it can evaluate the optical performance of completely unseen designs with high accuracy. Most importantly, it executes these predictions orders of magnitude faster than a standard physics simulation, enabling the near-instantaneous screening of potential design candidates [2, 5].

The Inverse Problem: Generative Networks for Novel Design
While rapid prediction is vital, the ultimate engineering goal is to automatically discover entirely new structures that satisfy specific optical requirements—a challenge known as the "inverse problem". To solve this, we employ generative neural networks. Instead of just calculating an output from a given input, generative models study our dataset to learn the underlying statistical rules and features of high-performing designs. By mastering this "recipe" for success, the generative network can synthesize multiple completely novel, fabricable design candidates tailored to exhibit specific target properties, such as a substantial optical band gap. This creates a direct, high-throughput pathway from a desired optical function to a viable physical blueprint [7, 10, 5].

However, while generative models provide a powerful approach for exploring the design

space, they are typically not explicitly conditioned on precise target spectra. In many practical applications, however, the desired optical response—such as the position and width of a stopband—is specified in advance, while material parameters remain fixed. This motivates the development of alternative inverse design strategies that directly optimize geometries for predefined spectral targets. In particular, we explore a latent-space-based optimization approach that combines a learned forward surrogate model with a structured representation of the geometry space [7, 5].

In addition to the classical unit-cell setting, we also consider more realistic photonic-crystal waveguide configurations. In this case, each design is represented by a low-resolution 2D refractive-index map, and the associated optical response includes transmission, reflection, and scattering losses. This setting introduces additional challenges, including limited data availability, incomplete labels, and increased physical complexity. It allows us to evaluate how well data-driven models generalize from idealized scenarios to more realistic design tasks [5].

To address the heightened physical complexity of these realistic waveguide configurations, we must also confront a fundamental mathematical hurdle: the ill-posed "one-to-many" mapping problem. Because wave physics permits multiple vastly different physical structures to produce the exact same optical response, standard deterministic regression models fail by mathematically averaging these conflicting shapes into blurred, non-functional predictions. To explicitly overcome this, we employ a conditional Denoising Diffusion Probabilistic Model (DDPM) framework. By reframing the generation of waveguide geometries as an iterative, stochastic denoising process, the DDPM explores the multimodal design space to synthesize distinct, valid structural profiles. Critically, by integrating advanced conditioning mechanisms and strict spatial regularizations, we force the diffusion model to simultaneously satisfy precise target reflection spectra while maintaining the topological continuity required for physical cleanroom fabrication [3, 4, 9].

Based on these considerations, this project addresses the following research question:

How effectively can data-driven neural network models be used to predict and design photonic crystal structures under realistic constraints?

More specifically, we investigate:

(i) To what extent can neural networks predict application-relevant spectral properties (such as bandgap edges or filter windows) from geometry?

(ii) How well do models trained on idealized 2D data generalize to more realistic device settings with finite structures and fabrication constraints?

(iii) Can inverse design methods reliably generate geometries that match predefined spectral targets?

(iv) How effectively can conditional diffusion models (DDPMs) overcome the "one-to-many" averaging failure in inverse design to yield physically stable and manufacturable geometries?

2 Background

Photonic Crystals (PhCs) are materials with a periodically varying permittivity $\varepsilon(r)$ that create a "photonic band gap," which allows certain light frequencies to pass while blocking others. Much like how semiconductors control the flow of electrons, PhCs offer unprecedented control over the propagation of light at the nanoscale. This unique capability makes them highly important for modern optics, enabling the tight confinement, guiding, and filtering of photons with near-perfect efficiency. Consequently, PhCs are widely used in advanced real-world applications, including high-speed telecommunication filters, low-loss waveguides, highly reflective mirrors, optical sensors, and integrated photonic circuits.

In the context of designing these devices and training our machine learning models, several parameters are critically important. The fundamental material parameters are the discrete refractive indices (permittivity) of the core and background materials. However, the exact optical behavior is primarily governed by the device's physical geometry—specifically, the 2D spatial shape of the unit cell, the transverse width profile, and the number of cascaded periods. These geometric input parameters directly determine the critical optical performance metrics, such as the position and size of the bandgap, as well as the wavelength-dependent reflection (S11), transmission (S12), and scattering loss spectra.

While the dataset provided by [2] served as a foundational resource, its utility was limited by discrepancies in geometric modeling and material selection relative to our fabrication goals. To overcome these design-space limitations, we curated a secondary PhC dataset. This custom data ensures that the training model accounts for the physical realities and material constraints essential to our project's objectives.

2.1 Data 1: Paper PhC data

Data source : https://github.com/clott3/PhC-2D-sq?tab=readme-ov-file.

Introduction of the Dataset (Qinlin)

It includes 20,000 samples. These samples are two-dimensional permittivity profiles with two different values. The permittivity values describe the optical properties of the materials. In particular, they determine the speed at which light propagates through the material. For each unit cell, the photonic crystal band structure is computed using the free

MIT Photonic Bands (MPB) software [6]. The band structure indicates whether certain frequencies of light can propagate through the photonic crystal or not. Figure 2.1 shows what the data look like. The first row presents the permittivity profile of a single PhC unit cell, which is used to train the predictor and generator. The second row shows the first six band structures obtained from the MPB simulation. In our project, we use 1052 samples that have a finite TM band gap. Instead of studying the full band structure, we focus on the bandgap size between the first and second bands, measured in units of c / a, as well as the gap-to-midgap ratio. These are two quantities that are important for practical PhC design.

Predictive Model for Bandgap-Window Estimation (Ramtin)

The quantities used in our forward prediction task are derived from the first two TM bands. Instead of modeling the entire band structure, we focus on the first TM bandgap, whose lower edge is defined as the maximum frequency of band 1 over the sampled k-points and whose upper edge is defined as the minimum frequency of band 2. From these two values, one can compute the gap width and the relative gap-to-midgap ratio, which are more directly relevant for device-oriented photonic crystal design than the full dispersion relation. This simplified representation is also consistent with the project goal of predicting practical "filter-window" quantities rather than reconstructing all simulated bands. In this sense, our task keeps the same geometric input as the Christensen dataset [2] but replaces the high-dimensional full-band output with a compact bandgap description that is easier to interpret and more closely tied to design specifications.

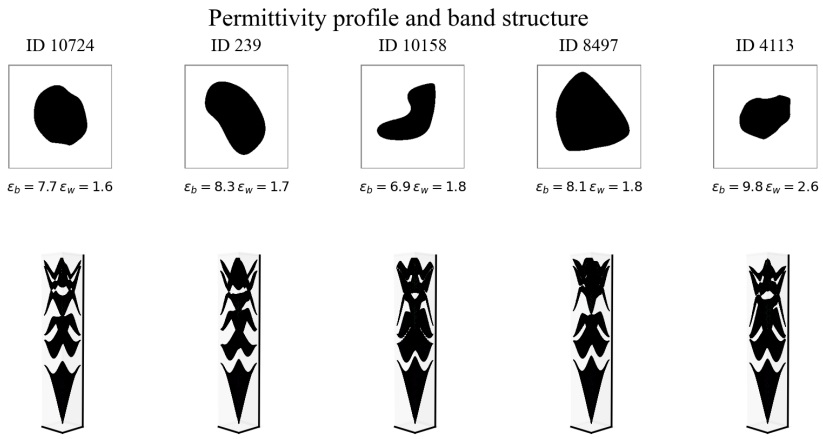

Figure 2.1: Paper's photonic crystal data set. First row: the dielectric function on the unit cell, as input to neural network. Each consists of smooth centered inclusion of permittivity ε_b and background permittivity ε_w, both are drawn uniformly from the range [1,10]. Second row: The band gap of the PhC above, includes first 6 bands.

2.2 Data 2: New PhC data (Qinlin)

The permittivity profiles presented in the paper provides a robust computational test bed, but they often rely on idealized parameters that are difficult to replicate in a physical cleanroom environment. Below are limitations regarding practical fabrication:

- **Fabrication Feasibility and Discrete Permittivity:** Existing models often draw permittivity values (ϵ) uniformly from a continuous range. In practical fabrication, materials (wafers) possess fixed, discrete permittivity values. Designs generated with arbitrary permittivity values are often unfeasible as they lack naturally occurring material counterparts or established fabrication recipes, rendering the optimization results effectively unusable in a cleanroom environment.
- **Dimensional Discrepancies:** High-throughput datasets are frequently restricted to 2D square lattices to maintain computational efficiency. However, physical PhC devices are inherently 3D. Neglecting the vertical dimension ignores critical thickness effects, which can cause significant deviations between simulated results and actual experimental performance.
- **Metric Realignment for Device Applications:** While existing research focuses heavily on the relationship between geometry and band structure or band gaps,

applying PhC structures as functional devices requires the consideration of additional physical properties. For applications such as PhC mirrors, metrics like insertion loss, coupling coefficients, and reflection spectra are more indicative of performance than the band gap and ratio alone.

To bridge the gap between theoretical machine learning models and practical engineering, this project develops a fabrication-focused dataset for Photonic Crystal (PhC) mirror design. We constrain the material parameters to industry-standard values: a core refractive index of $n = 1.99$ (Silicon Nitride, Si_3N_4) and a background index of $n = 1.44$ (Silicon Dioxide, SiO_2). Using commercial Ansys Lumerical FDTD software [1], we perform 3D geometric optimization to evaluate performance via reflection and loss spectra within the 1500–1600 nm telecommunications window.

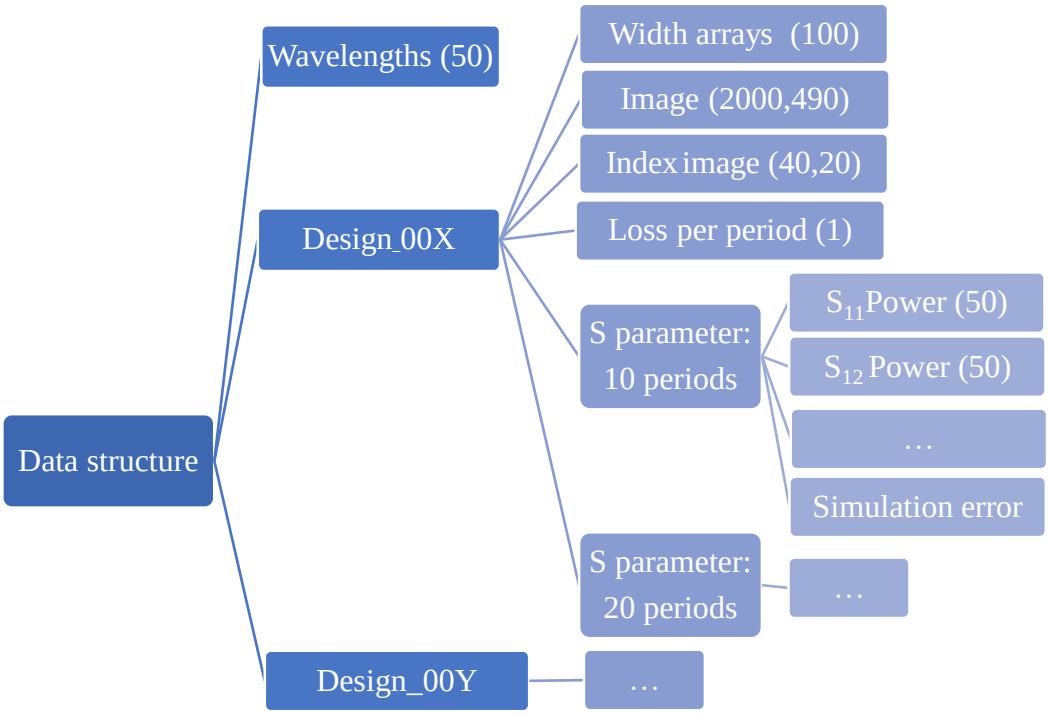

Figure 2.2: Tree diagram illustrating the dataset structure for a design group and its S-parameter subgroups.

The dataset structure, illustrated in 2.2, comprises 665 unique unit cell designs. Each design is y-symmetric and defined by a width array that describes how the y-directional width of the structure varies along the x-axis. The resulting morphology is shown in 2.3(a) (left). To maintain consistency with the simulation's discretization, the index images are down-sampled based on the x and y mesh resolution. 2.3(a) (right) displays the software's representation of the design, where the color bar denotes the effective refractive index.

The simulation configuration is depicted in 2.3(c). The PhC mirror with 10 periods

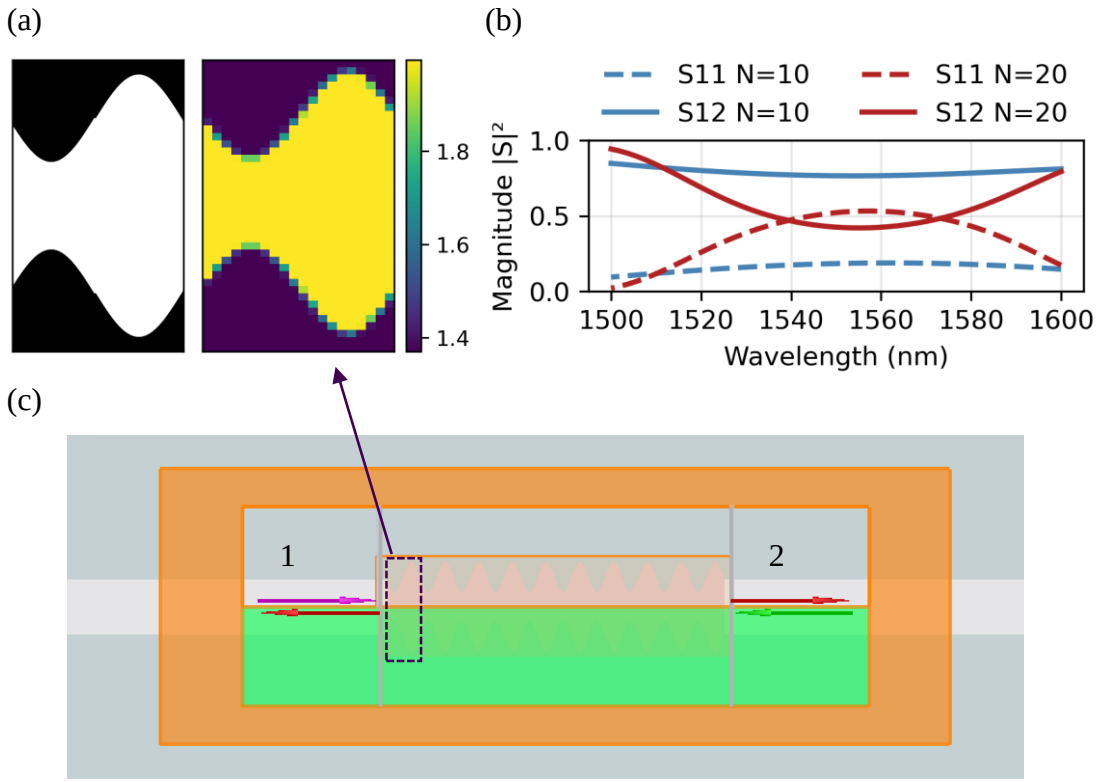

Figure 2.3: Geometry design and simulation data.

(highlighted in pink) is centered within the FDTD solver region (indicated by the orange box). Ports 1 and 2 act as both sources and monitors, capable of simultaneously emitting and detecting light. Each sample in the dataset includes two subsets: S-parameter power profiles simulated for different numbers of cascaded unit cells (periods). Specifically, S_{11} represents the power reflected back to Port 1 (reflectivity), while S_{12} denotes the power transmitted from Port 1 to Port 2 (transmissivity). As shown in 2.3(b), a PhC mirror with 20 periods (red dashed line) exhibits significantly higher reflectivity than the 10-period version (blue dashed line). Consistent with the principle of power conservation ($R + T \leq 1$), the increased reflection in the 20-period design results in a corresponding reduction in transmission.

Data Representation and Preprocessing

Each sample consists of a geometry vector $x \in \mathbb{R}^{100}$ and a corresponding transmission spectrum $y \in \mathbb{R}^{50}$. Designs containing NaN values are removed to ensure data consistency. For consistency across tasks, both geometry and spectral data are standardized per feature. The dataset is split into training (70%), validation (15%), and test (15%) sets using a fixed random seed.
Within this work, the dataset is used in two complementary representations. For forward prediction tasks, designs are represented as 2D refractive-index images, while for surrogate modeling and inverse design, geometries are represented as 1D width arrays.

2D Image Problem Setup and Data Representation

Our work specializes in applying the general machine-learning framework for photonics to a waveguide-device dataset in which each sample consists of a refractive-index image, a device length measured in number of periods, and wavelength-dependent optical responses. Concretely, each design is represented by an image $X \in \mathbb{R}^{40\times 20}$, where each pixel stores the local refractive index, and by a period number $n \in \{10, 20\}$. The main prediction targets are the reflection spectrum $S_{11} \in \mathbb{R}^{50}$, the transmission spectrum $S_{12} \in \mathbb{R}^{50}$, the scattering-loss spectrum $L \in \mathbb{R}^{50}$, and the scalar average loss $\bar{L} \in \mathbb{R}$. The wavelength axis contains 50 points in the telecom range from 1600 nm to 1500 nm. In order to keep the input scale consistent, each image was normalized as

$$\tilde{X} = \frac{X - \min(X)}{\max(X) - \min(X) + \varepsilon},$$

where $\tilde{X}$ is the normalized image and ε is a small constant used for numerical stability. The scalar input n was also normalized before being passed into the models:

$$\tilde{n} = \frac{n - 10}{10},$$

so that $n = 10$ maps to 0 and $n = 20$ maps to 1.

To ensure a consistent narrative, we have integrated the theoretical background and the architecture of the networks directly into Methods section.

3 Methods

3.1 Forward Predictive Modeling (Mapping Geometry to Spectra)

3.1.1 Predictive Model for Bandgap-Window Estimation (Ramtin)

Problem Formulation

For the predictive component, we specialize the general forward-modeling idea of Christensen et al.[2] to a lower-dimensional and more application-oriented setting. Instead of predicting the full photonic band structure over the sampled Brillouin zone, the goal here is to estimate compact descriptors of the first TM bandgap from the 2D unit-cell geometry. Concretely, the model predicts three outputs: a binary indicator of whether a substantial first TM gap is present, and the lower and upper edges of that gap. This formulation follows the project motivation that practical photonic design often depends more directly on the location and width of the stopband than on the detailed shape of all dispersion curves. The input is the unit-cell permittivity profile stored in the HDF5 dataset under `unitcell/epsilon`, represented as a single-channel image.

The regression targets are derived from the stored eigenfrequencies of the first two TM bands. Let $\omega_1(k)$ and $\omega_2(k)$ denote the first and second band frequencies at sampled wave vector k. The lower and upper edges of the first bandgap are defined as

$$\omega_{\text{low}} = \max_k \omega_1(k), \qquad \omega_{\text{high}} = \min_k \omega_2(k).$$

From these quantities, the relative gap-to-midgap ratio is computed as

$$g = \frac{\max\left(0, \omega_{\text{high}} - \omega_{\text{low}}\right)}{\frac{1}{2}\left(\omega_{\text{high}} + \omega_{\text{low}}\right)}.$$

A sample is labeled as gap-positive whenever $g \geq 0.05$, following the threshold used to identify substantial TM bandgaps. When available, the dataset field `bandgap\_1` is used for robust extraction of the stored gap ratio, while the edge values ω_{low} and ω_{high} are always computed from `eigfreqs` to ensure consistent regression targets. This use of compact bandgap quantities rather than the full band diagram is the main methodological adaptation introduced in this work.

Model Architecture

The predictive model, denoted GapNet in the implementation, is a convolutional neural network with a shared feature extractor and two task-specific output heads. The encoder consists of four convolutional blocks, each composed of convolution, batch normalization, ReLU activation, and max pooling. The resulting feature representation is reduced by adaptive average pooling and passed through a shared fully connected layer. From this latent representation, the network branches into a classification head that outputs a single logit for gap/no-gap prediction and a regression head that outputs the continuous pair $(\omega_{\text{low}}, \omega_{\text{high}})$.

This shared-trunk design allows the model to learn geometric features that are useful for both tasks while remaining substantially simpler than an architecture intended for full band-diagram reconstruction. In the context of the project requirements, the implemented design therefore represents a single focused CNN baseline rather than a broad architecture comparison. The main comparison within this part is instead conceptual: full band prediction from the original paper versus compact bandgap-window prediction in the present work, and internally, binary gap classification versus continuous edge regression.

Training Objective and Evaluation

Training is formulated as a multi-task learning problem. For the classification branch, we use weighted binary cross-entropy with logits,

$$L_{\text{cls}} = \text{BCEWithLogits}(z, y),$$

where z is the predicted logit and $y \in \{0, 1\}$ is the binary gap label. Binary cross-entropy with logits combines a sigmoid activation with binary cross-entropy in a numerically stable form, so that the model directly learns to separate gap-positive from gap-negative samples. The positive-class weight is chosen from the class distribution of the training set in order to reduce the effect of class imbalance.

For the regression branch, we use a masked SmoothL1 loss,

$$L_{\text{reg}} = \frac{1}{N_+} \sum_{i \in P} \text{SmoothL1}(\hat{\omega}_i, \omega_i) \,,$$

where i indexes the training samples, P denotes the set of positive-gap samples, $N_+ = |P|$, $\hat{\omega}_i = (\hat{\omega}_{\text{low},i}, \hat{\omega}_{\text{high},i})$ is the predicted edge pair, and $\omega_i = (\omega_{\text{low},i}, \omega_{\text{high},i})$ is the corresponding target pair. SmoothL1 loss behaves quadratically for small errors and linearly for large errors, making it less sensitive to outliers than a purely quadratic loss while still encouraging accurate regression. Masking is important because band-edge regression is only physically meaningful when a substantial gap exists. The total loss is then given by

$$L = L_{\text{cls}} + \lambda_{\text{reg}} L_{\text{reg}},$$

with $\lambda_{\text{reg}} = 1$ in the final implementation. This masked regression formulation is another idea added specifically in this work to better match the physics of the task.

Model optimization is performed using AdamW together with a learning-rate scheduler and early stopping based on the validation regression score. Since the main objective of the model is accurate band-edge prediction, model selection is based on regression performance on the validation set rather than classification accuracy alone. The classification threshold is selected on the validation set by maximizing the F1 score and then fixed for evaluation on the test set. For classification, performance is assessed using accuracy, precision, recall, specificity, balanced accuracy, and F1 score. For regression, performance is measured using the mean absolute error of ω_{low} and ω_{high} on true positive samples. Success is therefore defined primarily by the ability to predict the first bandgap window with low regression error; the binary classifier is treated as an auxiliary component that is useful if it improves gap/no-gap discrimination, but not as the main objective.

3.1.2 2D Image Forward Models (Ramtin)

For the forward-prediction setting, we implemented two closely related convolutional architectures. In both cases, the normalized image $\tilde{X}$ is processed by a Conv2D encoder $E_{\text{img}}(\tilde{X})$, while the normalized scalar period number $\tilde{n}$ is processed by a small multilayer perceptron $E_n(\tilde{n})$. Their latent representations are concatenated and passed to a shared prediction block:

$$h = f\big(E_{\text{img}}(\tilde{X}),\ E_n(\tilde{n})\big)\,,$$

where h denotes the fused latent representation and $f(\cdot)$ denotes the shared fully connected layers. The first model predicts reflection and transmission. Instead of predicting both outputs independently, we used a more physics-aware parameterization: the network predicts the reflection spectrum $\hat{S}_{11}$ and a nonnegative residual term $\hat{R}$, then defines

$$\hat{S}_{12} = 1 - \hat{S}_{11} - \hat{R}.$$

Here, $\hat{S}_{11}$ is the predicted reflection spectrum, $\hat{S}_{12}$ is the predicted transmission spectrum, and $\hat{R}$ represents the remaining power not attributed to transmission, for example radiation or other losses. This formulation encourages physically meaningful behavior by making it easier to satisfy $\hat{S}_{11} + \hat{S}_{12} \leq 1$ and by coupling the trends of reflection and transmission. The second forward model predicts $\hat{S}_{11}$, the loss spectrum $\hat{L}$, and the scalar average loss $\hat{\bar{L}}$ through separate output heads built on the same shared latent representation. This makes the architecture a multi-head predictor, where one head focuses on reflection, one on wavelength-dependent loss, and one on average loss.

The training losses were specialized to the properties of the dataset. Since each simulation also provides a reciprocity-based simulation error, each sample was assigned a weight

$$w_i = \frac{1}{10^{-4} + e_i},$$

where w_i is the weight for sample i and e_i is its simulation error. This gives slightly lower influence to less reliable simulations. For the reflection/transmission predictor, the main regression loss is a weighted mean squared error,

$$L_{\text{fit}} = \frac{1}{B} \sum_{i=1}^{B} w_i \, \frac{1}{100} \, \|\hat{y}_i - y_i\|_2^2 \,,$$

where B is the batch size, $y_i = [S_{11,i};\, S_{12,i}] \in \mathbb{R}^{100}$ is the true concatenated spectrum of sample i, and $\hat{y}_i = [\hat{S}_{11,i};\, \hat{S}_{12,i}]$ is the prediction. To encourage smoother spectra, we also used a second-difference regularizer

$$L_{\text{smooth}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{T-2} \sum_{t=2}^{T-1} \left(\hat{s}_{i,t+1} - 2\hat{s}_{i,t} + \hat{s}_{i,t-1}\right)^2 ,$$

where $T = 50$ is the number of wavelength points and $\hat{s}_{i,t}$ denotes a predicted spectral value at wavelength index t. In addition, we used a penalty for violating energy consistency,

$$L_{\text{energy}} = \frac{1}{BT} \sum_{i=1}^{B} \sum_{t=1}^{T} \max\Big(0, \hat{S}_{11,i,t} + \hat{S}_{12,i,t} - 1\Big)^2 ,$$

and an opposite-trend penalty

$$L_{\text{opp}} = \frac{1}{B(T-1)} \sum_{i=1}^{B} \sum_{t=1}^{T-1} \left[\left(\hat{S}_{11,i,t+1} - \hat{S}_{11,i,t}\right) + \left(\hat{S}_{12,i,t+1} - \hat{S}_{12,i,t}\right)\right]^2 .$$

The total loss for this model is

$$L_{4A} = L_{\text{fit}} + \lambda_{\text{smooth}} L_{\text{smooth}} + \lambda_{\text{energy}} L_{\text{energy}} + \lambda_{\text{opp}} L_{\text{opp}},$$

where λ_{smooth}, λ_{energy}, and λ_{opp} are hyperparameters controlling the relative importance of smoothing, energy consistency, and opposite spectral trends.

For the reflection/loss predictor, the main complication is that the loss spectrum contains missing values for some wavelengths. Therefore, we used a masked weighted mean squared error:

$$L_{\text{loss-spec}} = \frac{1}{B} \sum_{i=1}^{B} w_i \, \frac{\sum_{t=1}^{T} m_{i,t} \left(\hat{L}_{i,t} - L_{i,t}\right)^2}{\sum_{t=1}^{T} m_{i,t}},$$

where $m_{i,t} \in \{0, 1\}$ is a validity mask indicating whether the loss label at wavelength t is available for sample i. The full loss for this model is

$$L_{5A} = \alpha \, L_{S_{11}} + \beta \, L_{\text{loss-spec}} + \gamma \, L_{\text{avg-loss}},$$

where $L_{S_{11}}$ is the weighted MSE on the reflection spectrum, $L_{\text{avg-loss}}$ is the weighted MSE on the scalar average loss, and α, β, and γ are task weights. In our final version, we reduced β to prevent the difficult full loss spectrum from dominating training and harming the much more reliable reflection prediction. This was one of our own additions motivated by empirical behavior in training.

For the forward-prediction tasks, success was defined in terms of both low numerical error and visually faithful spectra. The main quantitative metric was the mean absolute error (MAE) over wavelength points for S_{11}, S_{12}, and valid loss values:

$$\text{MAE} = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left| \hat{y}_{i,t} - y_{i,t} \right|,$$

where N is the number of test samples and $T = 50$ is the number of wavelength points. For masked quantities such as the loss spectrum, the average was computed only over valid target entries. For the scalar average loss, we report the absolute error between $\hat{\bar{L}}$ and $\bar{L}$. In addition to these numerical metrics, qualitative comparisons between predicted and true spectra were also used to assess whether the models captured the main spectral features and overall physical behavior.

3.1.3 Differentiable Forward Surrogate Model (Milan)

3.1.4 Data Preparation

The inverse design approach is based on the New-PhC dataset, which is described in Section 2.1. Here, preprocessed geometry–spectrum pairs are used as input for training the forward model, the autoencoder and the inverse optimization pipeline.
Each sample consists of a geometry vector $x \in \mathbb{R}^{100}$ and a corresponding transmission spectrum $y \in \mathbb{R}^{50}$. Designs containing NaN values are removed to ensure stable training. To improve convergence, both the input geometries and spectral outputs are standardized per feature by subtracting the dataset mean and dividing by the standard deviation. This normalization reduces scale differences and improves the conditioning of the optimization problem.
The dataset is randomly split into training (70%), validation (15%), and test (15%) sets using a fixed random seed to ensure reproducibility.

Surrogate Model Architecture

To replace computationally expensive physical simulations, we train a forward model that maps normalized geometry parameters $x \in \mathbb{R}^{100}$ to their corresponding transmission spectrum $y \in \mathbb{R}^{50}$. The model is implemented as a fully connected multi-layer perceptron (MLP) with three hidden layers of sizes 512, 512, and 256, each using ReLU activations. The output layer is linear with 50 neurons, producing the predicted spectrum $\hat{y}$. We apply

a dropout rate of 0.1 in the first two hidden layers to reduce overfitting.
We train the network using the Adam optimizer with a learning rate of 10^{-3} on normalized input and output data. As the primary loss function, we employ the SmoothL1 loss, which provides stable gradients and is less sensitive to outliers than the mean squared error.
To ensure physically plausible spectra, we include additional regularization terms derived from finite differences of the predicted spectrum. Let $d_i^{(1)} = \hat{y}_{i+1} - \hat{y}_i$ denote first-order differences. A smoothness term

$$\text{smooth}_1 = \mathbb{E}[(d_i^{(1)})^2]$$

penalizes large local variations.
To reduce oscillatory behaviour, a monotonicity term penalises changes in sign between consecutive differences. This is achieved by increasing the loss when neighbouring differences have opposite signs and are weighted by their magnitude.
Additionally, second-order differences, defined as $d_i^{(2)} = d_{i+1}^{(1)} - d_i^{(1)}$ are used to define a curvature penalty

$$\text{smooth}_2 = \mathbb{E}[(d_i^{(2)})^2],$$

which discourages sharp changes in curvature.

The total loss function is given by

$$L_{\text{forward}} = L_{\text{SmoothL1}}(\hat{y}, y) + \lambda_{\text{smooth}} \cdot \text{smooth}_1 + \lambda_{\text{mono}} \cdot \text{mono_loss} + \lambda_{\text{smooth2}} \cdot \text{smooth}_2.$$

In our experiments, we use $\lambda_{\text{smooth}} = 6.6$, $\lambda_{\text{mono}} = 12.4$, and $\lambda_{\text{smooth2}} = 2.2$. The model was trained for 500 epochs on the training split and evaluated on the validation set to monitor its generalization performance. The resulting forward model can then be used as a differentiable surrogate for the physical simulation in the subsequent inverse design step.

3.2 Inverse Generative Design (Mapping Spectra to Geometry)

3.2.1 Deterministic and cVAE Baselines

Implementation of the Deterministic Baseline (Qinlin)

The conventional machine learning methodology for solving such mappings relies on deterministic supervised regression, most commonly utilizing the Multi-Layer Perceptron (MLP) [10]. An MLP is a feed-forward artificial neural network consisting of sequential layers of interconnected neurons, typically optimized using a continuous distance metric such as Mean Squared Error (MSE). However, when an MSE-optimized deterministic network is applied to highly multimodal data, it encounters a severe mathematical limitation.

Faced with divergent, equally valid physical solutions for a single input condition, the network minimizes its global error by predicting the geometric average of those solutions. In the context of physical design, this statistical regression to the mean produces blurred, compromised structures that fail to satisfy the actual physical constraints of any individual valid design.

To explicitly establish a benchmark for the averaging failure inherent to deterministic models, we implemented our first network variant: an MLP baseline. This network processes the 52-dimensional condition vector through a deep sequence of fully connected linear layers. The architecture expands from 256 to 512 hidden channels, utilizing Gaussian Error Linear Unit (GELU) activations and Dropout regularization ($p = 0.2$) to mitigate overfitting. The final layer projects the features directly into the 100-point spatial array. The network is trained using the Adam optimizer with a direct MSE loss function evaluated against the ground-truth 1D geometries over 150 epochs.

3.2.2 2D Image Inverse Models (Ramtin)

For the inverse-design setting, we compared two variants. The first is a deterministic baseline that maps a full target spectrum back to an image and period number. Its input is the concatenated spectral condition

$$c = [S_{11};\, S_{12}] \in \mathbb{R}^{100},$$

and the model outputs a predicted image $\hat{X} \in \mathbb{R}^{40\times 20}$ and a predicted normalized period number $\hat{n} \in [0, 1]$. The loss is

$$L_{4B} = L_{\text{img}} + 0.2\, L_n,$$

with

$$L_{\text{img}} = \frac{1}{B}\sum_{i=1}^{B} \left\|\hat{X}_i - X_i\right\|_2^2, \qquad L_n = \frac{1}{B}\sum_{i=1}^{B} (\hat{n}_i - \tilde{n}_i)^2 .$$

The second inverse-design variant addresses the more underdetermined case in which the target is only a single reflection value and a single loss value at a chosen wavelength λ^*. In this case, we use a conditional variational autoencoder (cVAE)[9]. The condition vector is

$$c = [S_{11}(\lambda^*),\, L(\lambda^*)] \in \mathbb{R}^2,$$

while the target output is the flattened image together with the normalized period number,

$$x = [\text{vec}(X),\, \tilde{n}] \in \mathbb{R}^{801}.$$

The encoder produces a mean vector $\mu(c, x)$ and a log-variance vector $\log \sigma^2(c, x)$, defining a latent Gaussian distribution. A latent sample is then generated through the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

where z is the latent code, σ is the standard deviation, $\odot$ denotes elementwise multiplication, and ϵ is random Gaussian noise. The decoder reconstructs $(\hat{X}, \hat{n})$ from (z, c). The cVAE loss is

$$L_{5B} = L_{\text{recon-img}} + 0.2\, L_{\text{recon-n}} + \beta_{\text{KL}}\, L_{\text{KL}},$$

where

$$L_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^{d} \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right)$$

is the Kullback–Leibler divergence between the learned latent distribution and the standard normal prior, and d is the latent dimension. This stochastic inverse model was chosen because inverse photonic design is generally one-to-many: several distinct geometries can satisfy similar target optical conditions, so a generative model is more appropriate than deterministic regression alone.

For the inverse-design tasks, pixelwise image accuracy alone was not considered sufficient, since multiple geometries may correspond to similar spectral responses. Therefore, in addition to visual comparisons between true and generated images, we used a forward-consistency check as the main practical evaluation criterion. In this procedure, a generated image is passed through the trained forward predictor, and the resulting spectrum is compared qualitatively with the target condition. This provides a more meaningful notion of success for inverse design in cases where an exact one-to-one reconstruction is not possible.

3.2.3 Latent-Space Optimization (Milan)

While previous sections focus on predictive and generative models, we now address the inverse design problem. In particular, we investigate whether latent-space-based optimization can reliably generate geometries that match predefined spectral targets.
Our goal is to solve an inverse design problem: for a given target spectrum y^*, we aim to find a geometry x such that the predicted spectrum $f_{\text{forward}}(x)$ closely reproduces y^*. Because direct physical simulations are computationally expensive, we use a trained forward model f_{forward} that approximates the mapping from normalized geometry parameters to normalized spectra.

As discussed in the introduction, the inverse mapping is ill-posed and non-unique. To address this, we introduce a latent-space-based optimization approach. Instead of directly

optimizing in the high-dimensional geometry space, we operate in a lower-dimensional latent space defined by a geometry autoencoder. This constrains the search to physically plausible designs.
Specifically, the geometry is parameterized as $x = D(z) + \Delta x$, where z is a latent representation and Δx is a small correction term. The latent vector z ensures that the solution remains close to the manifold of valid geometries, while Δx allows fine local adjustments beyond it.
The optimization is performed such that the predicted spectrum $f_{\text{forward}}(x)$ matches the target y^*. The objective is defined as the discrepancy between the predicted and target spectrum, and this discrepancy is minimized using gradient-based optimization methods.
The effectiveness of the inverse design approach is evaluated based on the ability to generate geometries whose predicted spectra closely match the target spectra. This is primarily measured using spectral mean squared error (MSE) to compare the predicted and target spectra.

Geometry Autoencoder

Direct optimization in geometry space often leads to physically implausible solutions. To constrain the search space, we employ a geometry autoencoder that defines a low-dimensional latent representation of valid structures. The autoencoder maps normalized geometry vectors $x \in \mathbb{R}^{100}$ to a latent representation $z \in \mathbb{R}^{16}$, and reconstructs them back to $\hat{x} \in \mathbb{R}^{100}$. The encoder consists of fully connected layers with dimensions $100 \to 128 \to 64 \to 16$, using ReLU activations. The decoder $D(\cdot)$ mirrors this structure with layers $16 \to 64 \to 128 \to 100$.
The autoencoder is trained exclusively using geometry data, with no spectral information. The model is trained using mean squared error (MSE) as the primary loss, comparing the input and the reconstruction. To regularize the latent space, an additional penalty term is introduced:

$$L_{\text{latent}} = \mathbb{E}[z^2]$$

This discourages large latent activations and restricts the effective search space. The total loss is given by

$$L = \|x - \hat{x}\|_2^2 + \lambda_{\text{latent}} \cdot L_{\text{latent}},$$

with $\lambda_{\text{latent}} = 10^{-3}$.
The model was trained for 130 epochs on the training set of the dataset, using a learning rate of 10^{-4}. After training, the autoencoder serves two purposes: (i) it acts as a prior, restricting generated geometries to the manifold of observed designs; and (ii) it provides a low-dimensional parameterization, enabling efficient optimization in latent space during inverse design.

Optimization in Latent Space

The goal of the inverse design step is to find a geometry $x \in \mathbb{R}^{100}$ such that the predicted spectrum $\hat{y} = f_{\text{forward}}(x)$ matches a given target spectrum $y_{\text{target}} \in \mathbb{R}^{50}$. Instead of directly optimizing in the high-dimensional geometry space, we perform the optimization in the latent space of the autoencoder, combined with a small adjustment in the original space. Specifically, the geometry is parameterized as:

$$x = D(z) + \Delta x,$$

where $z \in \mathbb{R}^{16}$ is a latent vector, $D(\cdot)$ denotes the decoder, and $\Delta x \in \mathbb{R}^{100}$ is an additive correction term. This formulation ensures that the solution remains close to the manifold of valid geometries learned by the autoencoder while still allowing local refinements.
While the autoencoder limits the solution space to realistic geometries, it can be too restrictive and may prevent an exact match to the target spectrum. The additional correction term Δx therefore introduces local flexibility, enabling fine adjustments beyond the learned manifold while still keeping the solution close to it.
Both z and Δx are treated as optimization variables. z is initialised using training data: a geometry sample, x_{init}, is taken from the training loader, encoded into a latent representation, z_{init}, and then perturbed with small Gaussian noise. The correction term Δx is initialized with small random values.
To improve robustness, we employed a multi-start strategy with $n_{\text{starts}} = 10$. For each initialization, the optimization is performed independently for up to 2,000 steps, and the best solution across all runs is selected based on the final loss. Optimization is carried out using Adam with a learning rate of $5 \cdot 10^{-3}$.
The objective function is defined as:

$$L_{\text{inv}} = \|f_{\text{forward}}(x) - y_{\text{target}}\|_2^2 + \lambda_z \mathbb{E}[z^2] + \lambda_{\Delta x} \mathbb{E}[\Delta x^2] + \lambda_{\text{mono}} \cdot L_{\text{mono}}(x)$$

The first term enforces spectral accuracy. The regularization terms with $\lambda_z = \lambda_{\Delta x} = 10^{-4}$ restrict the magnitude of the latent variables and corrections. This prevents the optimization from drifting too far away from the learned geometry manifold.
To promote smooth and manufacturable geometries, a monotonicity regularization is applied. Let $d_i^{\text{geom}} = x_{i+1} - x_i$ denote the first-order differences. The term $L_{\text{mono}}(x)$ penalizes changes in sign in consecutive differences, discouraging oscillatory behaviour and enforcing smoother geometric profiles. This term is weighted by $\lambda_{\text{mono}} = 0.02$.
In addition, after decoding and correction, the geometry is clamped to the interval $[0, 2]$, which acts as a simple box constraint to ensure numerical stability and prevent the occurrence of physically implausible parameter values. During optimization, gradient norm clipping with a threshold of 0.5 is applied to further improve numerical stability.
After all optimization runs have been completed, the geometry with the lowest loss is selected as the final design x_{opt}. The resulting spectrum is then evaluated using the forward model and compared with the target spectrum in order to assess the quality of the inverse design.

Despite this formulation, the inverse design problem remains challenging. The optimization landscape is highly non-convex, resulting in strong sensitivity to initialization and the presence of multiple local minima. Furthermore, the inverse mapping is inherently non-unique, meaning that different geometries can produce similar spectral responses. These challenges necessitate strategies such as multi-start optimization and regularization, which aim to improve robustness and guide the solution towards physically plausible designs.

3.2.4 Denoising Diffusion Probabilistic Models (Qinlin)

To overcome the averaging effect of deterministic networks, modern methodologies have shifted towards probabilistic generative modeling, specifically Denoising Diffusion Probabilistic Models (DDPMs) [3]. Instead of directly outputting a single rigid prediction, a DDPM learns to model the entire probability distribution of valid physical structures. This is achieved through a stochastic process governed by two Markov chains.

- The **forward diffusion** process systematically corrupts a valid sample from the data distribution by injecting Gaussian noise over a series of discrete timesteps, eventually degrading it into pure isotropic noise.
- The **reverse process**, parameterized by a neural network, is trained to progressively denoise this state step-by-step back into a valid sample.

The foundational neural architecture driving the reverse diffusion process is the U-Net [8]. A U-Net employs a symmetric encoder-decoder structure. The encoder applies convolutional layers to progressively down-sample the spatial dimensions of the input while expanding its feature channels, forcing the network to extract generalized, global representations in a central bottleneck. The decoder subsequently expands these features back to the original spatial resolution. To preserve the high-frequency local details lost during compression, skip connections directly link the encoder layers to their corresponding decoder layers.

For conditional generation tasks, external target data can be injected into the U-Net to steer the reverse process. Furthermore, sequence generation can be enhanced by embedding Multi-Head Self-Attention mechanisms [11], which correlate all elements in a sequence globally, bypassing the localized receptive fields of standard convolutions. Finally, target fidelity can be mathematically amplified using Classifier-Free Guidance (CFG) [4]. CFG trains the network to learn both conditional and unconditional distributions simultaneously. During inference, the algorithm extrapolates the prediction away from the unconditional prior and towards the conditioned target, effectively trading broad generative diversity for strict adherence to the desired physical parameters.

To resolve the one-to-many multimodality problem, we developed a Conditional 1D DDPM. Because physical manufacturing imposes strict geometric constraints, we progressively engineered three distinct variants of the U-Net architecture to ensure physical viability, adding our own structural constraints to the general diffusion framework.

DDPM V1:

Our initial generative implementation adapts a standard 1D Conditional U-Net. The noisy 100-point width array is processed through 1D convolutional residual blocks, while the 52-dimensional condition vector and the current diffusion timestep are embedded via linear layers and added directly into the feature maps. However, a fundamental architectural limitation arises when applying standard computer vision U-Nets to arbitrary physical arrays. Standard U-Nets apply symmetric pooling (e.g., stride-2 convolutions) that halves the spatial dimension at each encoder block. A 100-point array is mathematically indivisible by continuous powers of two ($100 \rightarrow 50 \rightarrow 25 \rightarrow 12.5$), which forces the network's deeper layers to rely on asymmetric spatial truncation. We hypothesized this indivisibility would compound interpolation errors during the 1000-step reverse sampling loop, manifesting as boundary artifacts.

DDPM V2 (Spatial Stabilization):

To address the structural mismatches of V1, we introduced our own architectural modifications to enforce boundary stability. First, rather than altering the physical dataset, we implemented a dynamic spatial padding mechanism at the U-Net input. Prior to the forward pass, the 100-point physical array is symmetrically padded to 128 points (a mathematically clean power of two, 2^7) using boundary replication (`mode='replicate'`) instead of zero-padding, which prevents the formation of artificial step-edges. Following the decoder, the tensor is symmetrically cropped back to 100 points.

Second, we injected a physical prior directly into the reverse Langevin sampling phase via a strict thermodynamic clamping mechanism. At the conclusion of every single denoising timestep, the entire synthesized coordinate array is mathematically clamped to the $[-1.0, 1.0]$ boundary. This algorithm explicitly forces the generative process to respect the maximum permissible physical footprint of the wafer, preventing the diverging coordinate variance often observed in unconstrained diffusion sampling.

DDPM v3 (High-Fidelity Optimization):

While V2 enforces physical manufacturability, generative models exploring broad multimodal distributions frequently suffer from "conditioning collapse"—generating a valid physical structure that ignores the specific optical target. To bridge the gap between generating *any* valid waveguide and generating the *optimal* waveguide for a specific S_{11} spectrum, we engineered Variant 3 by adding two advanced mechanisms.

First, recognizing that optical resonance is a holistic property of the entire optical cavity rather than a localized pixel feature, we embedded a Multi-Head Self-Attention block directly into the U-Net bottleneck. This expands the network's receptive field from local convolutional kernels to the entire 1D sequence, enabling the network to globally correlate specific target frequencies to distant geometric ripples.

Second, we implemented Classifier-Free Guidance (CFG). During the training loop, the 52-dimensional condition vector is dynamically masked with zeros for a random 10% of the batch, forcing the model to simultaneously learn conditional and unconditional spatial distributions. During inference, the DDPM predicts the noise twice per step (with and without the target). We apply a guidance scale of $w = 3.0$ to calculate the vector difference between these predictions, aggressively extrapolating the sampling trajectory to forcefully align the generated geometry with the specific optical target.

Definition of Success and Evaluation Metrics

Success in this inverse design task is not defined by a single numerical metric, but by achieving a balance between quantitative target fidelity and qualitative physical manufacturability. We evaluate all models strictly on unseen test data to prevent data leakage.

Quantitative Metrics: The primary numerical metric is the unnormalized spatial Mean Squared Error (MSE) computed between the predicted 1D width array and the specific ground-truth structure paired with the target spectrum. A highly successful model should achieve a spatial MSE competitive with or lower than the deterministic MLP baseline. However, we explicitly acknowledge the "MSE paradox" inherent to generative physics: because multiple distinct geometries can satisfy the same spectrum, a slightly higher structural MSE in a DDPM does not necessarily indicate optical failure, provided the model has sampled a valid alternative physical mode.

Qualitative Metrics and Algorithmic Rendering: Because purely mathematical evaluation is insufficient for physical cleanroom design, our secondary metric evaluates the topological continuity of the output. To measure this, we developed a deterministic rendering algorithm that interpolates the 100-point 1D arrays into high-resolution 2000×490 binary 2D spatial masks. This matrix acts as a simulated lithographic blueprint of the device. Success is visually and topologically defined by the absence of high-frequency fabrication artifacts. The 2D boundaries must be smooth, continuously differentiable, and completely devoid of jagged edges, unphysical vertical step-cliffs, or terminal boundary collapse. Abrupt discontinuities induce catastrophic out-of-plane light scattering in real-world fabrication. Therefore, a model is considered fully successful only if it surpasses the deterministic baseline in quantitative accuracy while strictly maintaining continuous, lithography-friendly physical boundaries [5].

3.3 Project Planning

To tackle the computationally expensive challenge of nanophotonic inverse design, we organized our project as a cohesive, end-to-end machine learning pipeline rather than isolated tasks. By strategically allocating our computational resources across three interdependent phases—data generation, forward evaluation, and parallel inverse design—we maximized efficiency and ensured synergistic research.

3.3.1 Individual Contributions

Qinlin grounded the project in physical reality by simulating a PhC waveguide dataset. She then tackled the inverse problem using Denoising Diffusion Probabilistic Models (DDPMs), progressively engineering architectural constraints to ensure the generated optical mirrors were continuous and fabrication-friendly.

Ramtin solved the critical rapid-evaluation bottleneck. He developed robust Convolutional Neural Networks (CNNs) to solve the forward problem, instantly predicting bandgaps, reflection, transmission, and loss spectra directly from physical geometries. Furthermore, he established baseline inverse models (MLP and cVAE[9]) to formally map the ill-posed, "one-to-many" nature of photonic design.

Milan pursued an alternative inverse design strategy via latent-space optimization. He trained a geometry autoencoder to compress valid shapes into a mathematically constrained latent space. Using a custom forward surrogate model, he applied gradient-based optimization to meticulously tailor generated geometries to match exact target spectra.

3.3.2 Effort Estimation and Subgoals

Qinlin:

- **Phase 1: Dataset Generation via FDTD Simulations**
  - *Subgoal:* Build a custom dataset under strict fabrication constraints.
  - *Effort:* This phase required significantly more effort than anticipated. While a 10-period simulation concluded in 1 min, a 50-period structure required 20 min. This non-linear scaling forced us to reduce the total periodicity (10, 20) to maintain the project timeline.
- **Phase 2: Baseline Model Implementation**
  - *Subgoal:* Deploy a deterministic MLP baseline and the initial DDPM.
  - *Effort:* While implementation proceeded faster than anticipated, the baseline's failure to produce functional geometries validated the shift toward generative architectures.
- **Phase 3: Generative Evolution (v1 to v3)**
  - *Subgoal:* Refine the model to bridge the "Simulation-to-Reality" (Sim2Real) gap.
  - *Effort:* This required a massive, unexpected engineering pivot. Initial models (v1) exhibited "vertical cliffs" caused by interpolation errors in the 100-point array. After research, it is resolved (v2) by implementing spatial padding to 128 points and strict $[-1.0, 1.0]$ boundary clamping. While v3 achieved

a lower MSE, v2 proved superior for fabrication; its natural smoothing bias created adiabatic tapers that respect lithographic limits.

- **Phase 4: Final Validation and Cross-Verification (not done)**
- *Subgoal:* Verify all results using both simulation and experimental data.
- *Reason:* Due to extensive troubleshooting during model training, we were unable to cross-verify the generative design and instead relied solely on existing data.

Ramtin:

- **Phase 1: Predictive Model for Bandgap-Window Estimation**
  * *Subgoal:* Define bandgap-edge labels from the 2D data, implement a CNN baseline for predicting compact filter-window quantities, and evaluate both regression and classification performance.
  * *Effort:* This phase was feasible within the project scope because the prediction target was reduced to a compact output setting. However, the main challenge was not the CNN implementation itself, but the construction of reliable labels and the design of a meaningful evaluation procedure. As a result, more effort went into preprocessing and evaluation setup than into the actual network coding.
- **Phase 2: 2D Image Forward Models**
  * *Subgoal:* Predict the full reflection and transmission spectra from 2D refractive-index images and period number, and additionally predict reflection together with scattering-loss quantities from the same inputs.
  * *Effort:* This phase required substantial effort, mainly to make the predictors stable and physically meaningful. A major difficulty was handling incomplete loss labels while also balancing the easier reflection target against the more difficult loss-spectrum target. Considerable experimentation was needed to obtain robust training behavior. Overall, this part was completed successfully and became the strongest component of the notebook.
- **Phase 3: 2D Image Inverse Models**
  * *Subgoal:* Reconstruct a plausible geometry and recommended period number from a target reflection/transmission spectrum, and generate candidate geometries from a single target reflection value and loss value at one wavelength. A further planned extension was to generate geometry directly from average reflection and average loss targets.
  * *Effort:* This phase required careful experimentation because the inverse mapping from spectra to geometry is fundamentally non-unique. In ad-

dition, stronger geometric constraints often made optimization harder instead of easier. The planned extension using average reflection and average loss targets was not completed. Most of the effort therefore focused on establishing stable baseline inverse models and evaluating them through forward-consistency checks rather than exact image reconstruction alone.

Milan:

- **Phase 1: Forward Surrogate Model Development**
  - *Subgoal:* Train a stable and accurate surrogate model mapping geometry to transmission spectra.
  - *Effort:* This phase required significantly more effort than initially expected. While the initial architecture design appeared straightforward, the development process was highly iterative rather than linear. Additional parameters and regularization terms were repeatedly introduced and adjusted during training.
    Achieving stable training proved challenging due to strong sensitivity to hyperparameters and loss weighting. Several configurations led to unstable or oscillatory predictions, so a large portion of time was spent on tuning and debugging rather than initial implementation.
- **Phase 2: Geometry Autoencoder**
  - *Subgoal:* Learn a compact latent representation of valid geometries to constrain the inverse design search space.
  - *Effort:* This phase was completed largely as planned and required moderate effort. The main challenge was an unexpected interaction between the autoencoder and the forward model. Although trained independently, the latent representation influenced the forward predictions and thus the inverse design results.
    This required careful control of the autoencoder training. Adding a latent regularization term helped stabilize the latent space and improve consistency.
- **Phase 3: Latent-Space Inverse Design**
  - *Subgoal:* Optimize geometries in latent space such that the predicted spectrum matches a given target.
  - *Effort:* Initially expected to be the most challenging part, this phase turned out to be more manageable once a stable forward model was available. The main effort here was spent on designing a robust optimization strategy, including multi-start initialization, regularization of latent variables, and enforcing smooth geometries. Without a reliable forward model, this step would not have been feasible.

3.3.3 Integration and Team Synergy

Our collaboration involved more than just a simple division of tasks; it also involved substantial knowledge transfer within the team. Qinlin, who had prior experience in photonic crystal design and simulation, played a pivotal role in helping Ramtin and Milan to understand the underlying physical principles and the structure of the dataset. This was crucial for correctly interpreting the data and preventing unrealistic modelling assumptions.

Based on this shared understanding, our individual contributions came together to form a tightly coupled system. Qinlin's centralised dataset provided a consistent foundation, while Ramtin's and Milan's forward models enabled the rapid evaluation of generated designs.

The inverse problem was addressed from multiple perspectives in parallel: Ramtin's stochastic cVAE [9], Milan's latent-space optimisation and Qinlin's diffusion-based approach. This parallel development enabled direct comparison of different methodologies and the identification of consistent patterns across them.

Notably, this collaborative setup enabled us to reveal the trade-off between mathematical accuracy and physical manufacturability — a finding that would have been challenging to attain using a single method.

4 Experiment and Results

4.1 Forward Prediction Results

4.1.1 Predictive Model for Bandgap-Window Estimation (Ramtin)

Experimental Setup

The experiments were conducted on the filtered Christensen TM dataset [2]. After filtering, the dataset contained 1052 valid samples, including 585 positive-gap and 467 negative-gap examples. A stratified split produced 842 training samples, 105 validation samples, and 105 test samples. The gap threshold was fixed at 0.05, corresponding to the criterion for substantial first TM gaps used in the code.

All runs used min-max normalization and the GapNet architecture. The final training setup used at most 80 epochs, a batch size of 128, a learning rate of 3×10^{-5}, regression weight $\lambda_{\text{reg}} = 1.0$, AdamW weight decay of 10^{-4}, and early stopping with patience 10. A ReduceLROnPlateau scheduler with factor 0.5 and patience 3 was also used.

For evaluation, the classification threshold was selected on the validation set by maximizing the F1 score and then applied to the test set. Regression performance was measured using MAE for ω_{low} and ω_{high} on true positive test samples. The notebook also generates a probability histogram, a confusion matrix, and scatter plots of true versus predicted band-edge values for qualitative analysis.

Results

The predictive model was evaluated as a compact surrogate for the first TM bandgap window. Using the threshold selected on the validation set, the final operating point on the test set was 0.40. At this threshold, the classifier achieved an F1 score of 0.7389, with recall 0.9831 and specificity 0.1304. The corresponding confusion matrix was

$$\begin{bmatrix} 6 & 40 \\ 1 & 58 \end{bmatrix}$$

where rows denote the true labels and columns denote the predicted labels.

The regression performance was evaluated on the 59 true positive test samples, since the band-edge targets are only physically meaningful when a substantial gap is present. The model achieved a mean absolute error of 0.060311 for the lower edge ω_{low} and 0.023366 for the upper edge ω_{high}.

The regression plots provide additional qualitative information. In Figure 4.1, which shows predicted versus true values for ω_{low}, the points follow the correct overall trend but exhibit noticeable spread around the diagonal. Figure 4.2 shows the corresponding results for ω_{high}, where the points also follow the correct trend and appear somewhat more concentrated around the diagonal.

Discussion

These results help answer the main research question of whether a CNN can predict practically relevant bandgap-window quantities directly from the unit-cell image. The answer is positive in a limited but meaningful sense: the regression results show that the model can recover useful information about the first TM gap from geometry alone, without reconstructing the full band diagram. This supports the overall goal of replacing a high-dimensional prediction target with a simpler and more application-oriented representation.

At the same time, the results show that the two prediction tasks are not equally successful. The regression component is the stronger part of the model. The scatter plots in Figures 4.1 and 4.2 show clear positive trends, which indicates that the network has learned meaningful correlations between unit-cell geometry and the band-edge values. However, the visible spread around the diagonal also shows that the regression accuracy is moderate rather than highly precise. In particular, the lower edge ω_{low} is more difficult to predict, whereas the tighter alignment in Figure 4.2 is consistent with the lower MAE for ω_{high}.

The classification component is clearly less convincing. The confusion matrix and the combination of recall and specificity show that the model identifies almost all positive-gap samples, but does so by overpredicting the positive class. This means that the current architecture is much better at estimating approximate bandgap boundaries than at reliably deciding whether a substantial gap exists in the first place. As a result, the model should be interpreted primarily as a proof-of-concept regressor for compact bandgap descriptors, rather than as a strong standalone gap detector.

Overall, these findings are consistent with the motivation behind this predictive model. Predicting compact spectral descriptors is indeed more manageable than reconstructing the full band structure, and the present results suggest that this simplification is useful. However, the current performance also shows that further work would be needed before such a model could serve as a robust design tool, especially if reliable gap/no-gap classification or integration with more realistic 3D

performance labels is required.

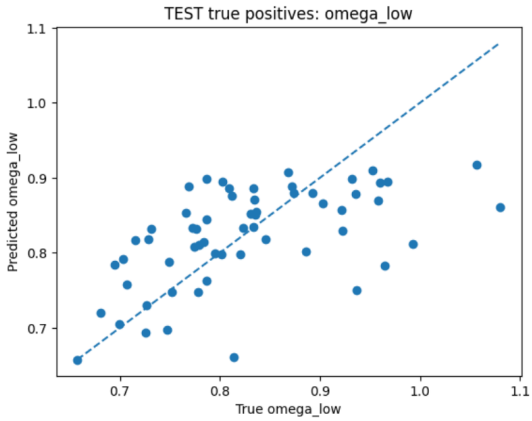

Figure 4.1: Predicted versus true lower edge ω_{low} of the first TM bandgap on true positive test samples.

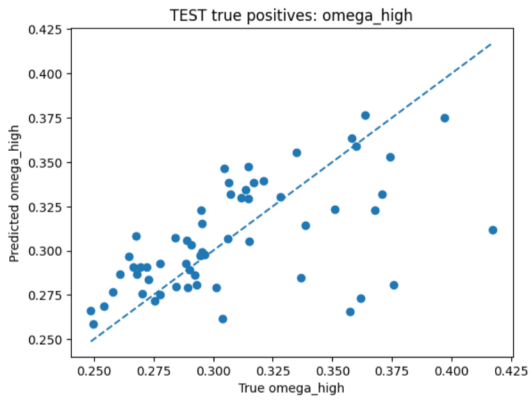

Figure 4.2: Predicted versus true upper edge ω_{high} of the first TM bandgap on true positive test samples.

4.1.2 2D Image Forward Models (Ramtin)

Experimental setup

The forward-prediction experiments were designed to evaluate how accurately optical responses can be predicted from geometry and period number. All forward models were implemented in PyTorch and trained on GPU when available. The models were trained with a batch size of 32 using the Adam optimizer with weight decay, and the learning rate for the forward predictors was set to 8×10^{-4}. In the final notebook version, the reflection–transmission predictor was trained for up to about 1000 epochs, while the reflection–loss predictor was trained for about 100 epochs. In both cases, the best-performing model was selected according to the validation loss. The data were always split at the design level rather than the sample level, so that the same geometry could not appear in both training and test sets under different period numbers.

For the forward models, the main quantitative metric was the test-set mean absolute error (MAE), supported by qualitative comparisons between predicted and true spectra. This setup was chosen to answer the forward research question systematically, namely whether compact neural networks can serve as accurate surrogate models for predicting the spectral response of photonic-crystal waveguides from their refractive-index images.

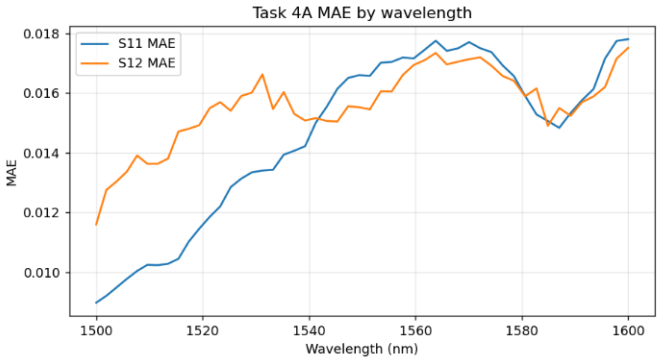

Figure 4.3: MAE by wavelength

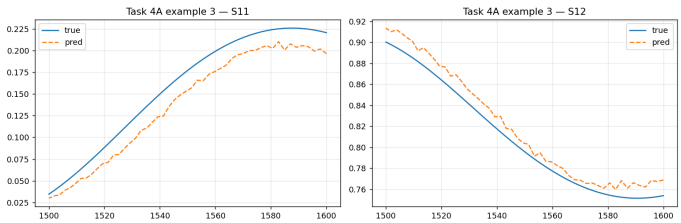

Figure 4.4: Representative true vs. predicted S_{11} and S_{12}

Results for reflection and transmission prediction

The strongest performance in the project was obtained for the reflection/transmission predictor. In the final version, the model achieved a test MAE of approximately 0.0150 overall, with 0.0145 for S_{11} and about 0.0155 for S_{12}. This indicates that the model learned the mapping from refractive-index image and period number to spectral response with good accuracy.

This conclusion is supported by both the quantitative and qualitative plots. Figure 4.3 shows the wavelength-wise MAE for S_{11} and S_{12}. The error remains low across the full wavelength range, which suggests that the predictor is capturing the structure of the full spectrum rather than only matching isolated points. Figure 4.4 shows one representative overlay of true and predicted S_{11} and S_{12}. In these plots, the predicted curves follow the true spectral shape closely and reproduce the main stop-band behavior. Finally, Figure 4.5 demonstrates that when period numbers larger than those seen during training are supplied, the predicted spectra remain bounded and physically reasonable rather than exploding numerically. Together, these results show that the forward surrogate is accurate, stable, and useful as a fast approximation of the underlying simulation.

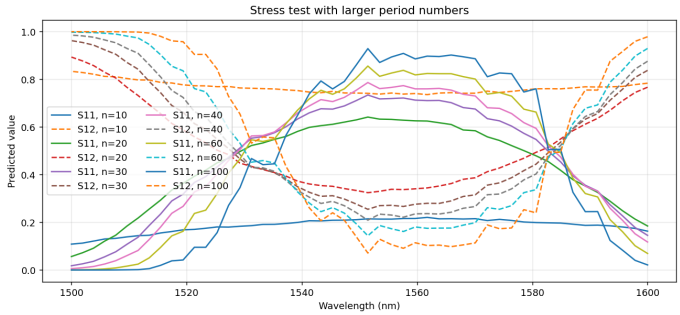

Figure 4.5: Robustness of the reflection–transmission predictor outside the training range

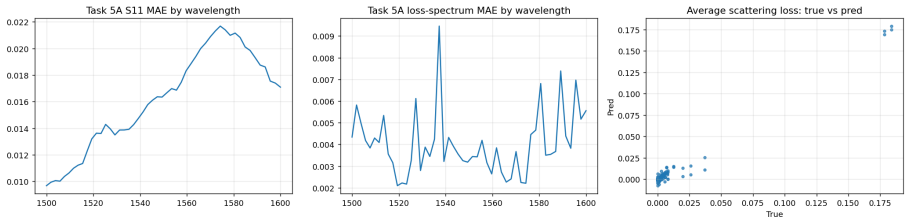

Figure 4.6: Error summary for the reflection–loss predictor: MAE of S_{11}, MAE of the loss spectrum, and true vs. predicted average scattering loss.

Results for reflection and loss prediction

The second forward model produced more mixed results. The reflection part remained strong, with a test MAE of roughly 0.0160 for S_{11}, while the scalar average-loss prediction also showed useful accuracy. By contrast, the wavelength-dependent loss-spectrum prediction was clearly more difficult and less stable. This suggests that the model is able to learn the dominant reflection behavior reliably, but that the full scattering-loss spectrum remains a harder target.

This difference is visible in the plots. Figure 4.6 summarizes the errors of this model. The first panel shows that the MAE for S_{11} stays relatively low across wavelength, while the second panel shows that the loss-spectrum error is less consistent. The third panel, which compares true and predicted average scattering loss, indicates that the scalar loss summary is easier to learn than the full wavelength-by-wavelength loss curve. Figure 4.7 shows one representative qualitative example. In this example, the predicted S_{11} follows the true trend and captures the main spectral structure, while the predicted loss spectrum is visibly noisier and less physically convincing. For this reason, reducing the weight of the loss-spectrum term during training was a reasonable choice: it improved the balance of the model and prevented the hardest target from degrading the more reliable reflection prediction. Overall, these experiments suggest that the model predicts reflection well and average loss reasonably well, but that the full scattering-loss spectrum remains the most challenging forward target.

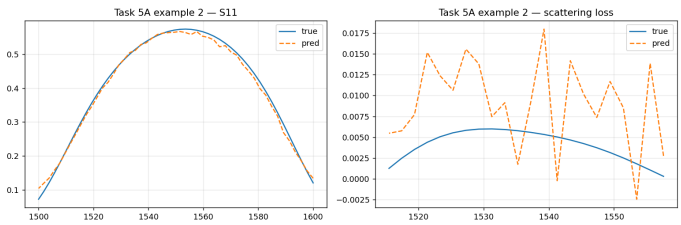

Figure 4.7: Representative true vs. predicted S_{11} and scattering-loss spectrum.

Discussion

The forward experiments show that the most successful part of the project is the surrogate modeling of optical response from geometry. In particular, the reflection–transmission predictor achieved low numerical error, visually accurate spectra, and stable behavior even under stress testing. This suggests that compact CNN-based models can provide an efficient approximation of expensive waveguide simulations. The reflection–loss predictor was also useful, but it made clear that the full scattering-loss spectrum is much harder to learn than either reflection or average loss. Therefore, the main conclusion from the forward experiments is that machine learning already works well here as a fast predictive tool, especially for spectral quantities that are stable and well represented in the dataset.

4.1.3 Surrogate Forward-Model (Milan)

Experimental Setup

The forward surrogate model was trained on the training split of the New-PhC dataset for 500 epochs. Model performance was monitored using the validation set and finally evaluated on the held-out test set.

To assess the predictive accuracy, the mean squared error (MSE) between predicted and ground-truth spectra was computed. In addition, representative examples of true and predicted spectra were visualized to qualitatively evaluate the model performance and to identify potential oscillatory artifacts.

Results of the Forward Model

The forward model shows a clear convergence behaviour during training (see Fig. 4.8). Both the training and validation losses decrease rapidly during the first epochs, before stabilizing at low values. The training loss falls from around 0.39 in the first epoch to 0.012 after 500 epochs, while the validation loss decreases from 0.38 to 0.017. The close agreement between the two curves indicates stable convergence without strong overfitting. On the held-out test set, the model achieves a loss of 0.028.

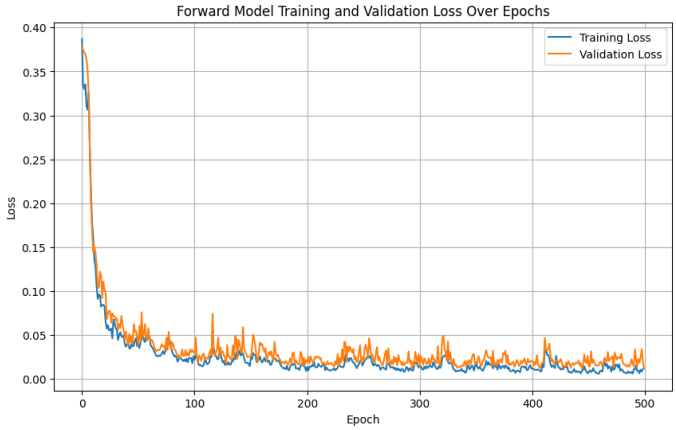

Figure 4.8: Training and validation loss of the forward model over 500 epochs. Both losses decrease rapidly in the initial phase and then gradually stabilize at low values, showing no significant gap between training and validation, which indicates stable convergence without strong overfitting.

To further characterize the predictions, we compared the statistical properties of the predicted spectra with the ground truth. The standard deviation of the predicted spectra (0.766) closely matches that of the true spectra (0.751), indicating that the model captures the overall variability of the data. The mean of the predictions (0.098) is slightly lower than the true mean (0.107).
The mean absolute error (MAE) is 0.101, with maximum pointwise deviations of up to 1.34. Larger errors primarily occur at individual wavelength points and in regions with stronger local gradients.

Figure 4.9 (True vs. Predicted Spectrum) shows a representative example. The overall shape of the spectrum, including the position and width of the main peak, is captured well. However, the predicted spectrum shows a slightly reduced amplitude, small local fluctuations and slightly increased noise compared to the true spectrum.

Discussion

The results show that the forward model can accurately predict the mapping from geometry to spectral response. The close agreement between the losses for the training, validation and test sets indicates that the model generalizes well and does not exhibit significant overfitting, despite the relatively small size of the dataset.
However, a systematic deviation between the predicted and true spectra can be observed. In particular, the predicted spectra tend to underestimate the amplitude of the true signal and appear smoother. This behaviour is consistent with the additional regularization terms introduced during training. The smoothness and monotonic-

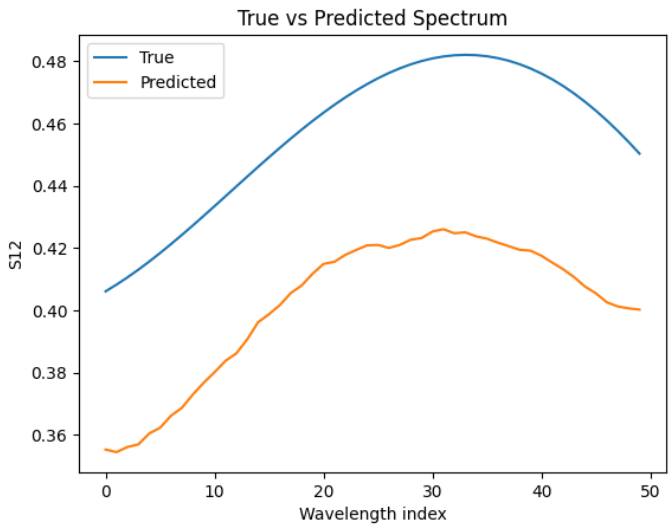

Figure 4.9: Comparison between the predicted and ground truth transmission spectrum for a test sample. The forward model captures the overall shape of the spectrum, but underestimates the amplitude and shows small local fluctuations.

ity constraints reduce oscillatory artefacts and enforce physically plausible spectral shapes, but at the cost of slightly reduced point-wise accuracy.
The relatively large maximum error at individual wavelength suggests that the model has difficulty capturing sharp local variations in the spectrum. This is to be expected, given that both the SmoothL1 loss and the regularisation terms penalise strong local changes, thereby biasing the model towards smoother predictions.
In summary, the forward model provides a stable and differentiable approximation of the underlying physical simulation. While it does not reproduce all fine-grained spectral details perfectly, it captures the global structure of the spectra sufficiently well to serve as a surrogate model for the inverse design task.

4.2 Inverse Design Results

4.2.1 2D Image Inverse Models (Ramtin)

Experimental setup

The inverse-design experiments were designed to evaluate whether useful candidate geometries can be recovered from desired optical targets. Two inverse settings were considered. In the first, a deterministic baseline model was trained to map a full target reflection/transmission spectrum back to a geometry and period number. In the second, a conditional variational autoencoder (cVAE) [9] was trained to generate candidate geometries from a single target reflection value and a single target loss

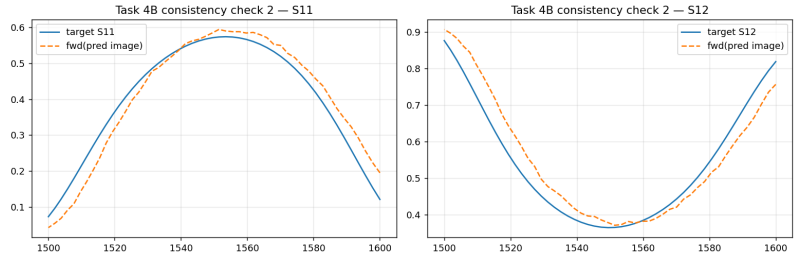

Figure 4.10: Forward-consistency check for inverse design from full spectra. The generated geometry is passed through the forward predictor and compared with the original target spectra.

value at a chosen wavelength. Both inverse models were implemented in PyTorch and trained on GPU when available. The deterministic inverse baseline and the cVAE used the Adam optimizer with learning rates on the order of 10^{-3}, a batch size of 32, and were trained for about 500 epochs in the final notebook version, with the best-performing model selected according to the validation loss.

For inverse design, numerical reconstruction loss was monitored during training, but this was not treated as the only evaluation criterion. Since inverse photonic design is not one-to-one, a generated geometry was considered useful if it produced the desired spectral behavior when evaluated by the trained forward predictor. For this reason, the main qualitative evaluation was based on generated-image inspection and forward-consistency checks.

Results for inverse design from full spectra

The inverse model conditioned on full reflection and transmission spectra was able to train stably and produce structured outputs, but its reconstructions were weaker than the forward results. The generated images often matched the coarse layout of the target geometry, yet they were blurrier than the true images and did not recover fine details exactly. This is not unexpected, since several different geometries can produce similar spectra.

For this reason, the most important analysis is not the image reconstruction itself but the forward-consistency check. Figure 4.10 shows the comparison between target spectra and the spectra obtained after passing the generated image back through the trained forward predictor. Even when the reconstructed geometry is not visually exact, the regenerated spectra often preserve the main trend of the target reflection and transmission curves. This indicates that the inverse model is still learning a meaningful mapping from spectral conditions to geometrical candidates. Therefore, the full-spectrum inverse model should be interpreted as a baseline inverse-design tool that produces spectrally plausible candidates, rather than as a perfect image-reconstruction model.

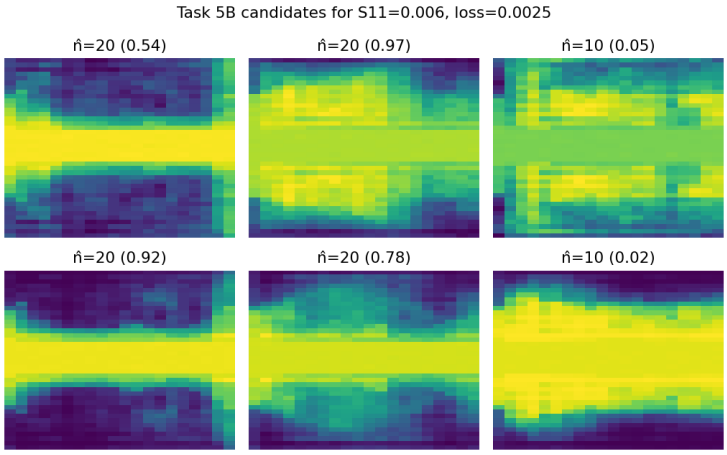

Figure 4.11: Multiple candidate geometries generated for the same single-wavelength target condition.

Results for inverse design from single-wavelength targets

The conditional variational autoencoder addressed the most underdetermined setting in the project: generating geometries from only one target reflection value and one target loss value at a chosen wavelength. In this case, exact reconstruction is not realistic, so the main objective was to generate several plausible candidates rather than a single exact answer.

This behavior is visible in Figure 4.11, where multiple distinct candidate geometries are generated for the same target condition. This result is important because it demonstrates the one-to-many nature of the inverse problem: different structures can satisfy similar optical requirements. The conditional variational autoencoder is therefore more appropriate than deterministic regression in this setting. Instead of forcing the model toward a single blurred average solution, it provides a small set of alternative candidate designs that can later be screened with the forward predictor or with full simulation. It is also worth noting that the number in parentheses is the raw predicted value $\hat{n}$ produced by the decoder. A value closer to 1 means the model is more confident that the correct period number is 20, while a value closer to 0 means it is more confident that the correct period number is 10. Values near 0.5 indicate uncertainty, since they lie close to the decision threshold between the two classes. Conceptually, this makes the conditional variational autoencoder the more suitable inverse model for single-wavelength design constraints.

Discussion

The inverse experiments gave more limited results than the forward ones, but they still provide useful conclusions. The deterministic inverse model was able to generate geometries whose spectral behavior remained plausible under forward checking,

even though the reconstructed images were not visually exact. This shows that some meaningful inverse mapping was learned, but also confirms that this task is much harder than forward prediction. The cVAE [9] provided a better conceptual match for the reduced single-wavelength problem, since it explicitly captured the one-to-many nature of inverse design by generating multiple candidate structures for the same target condition. The main conclusion from the inverse experiments is therefore more cautious: inverse photonic design is feasible at a baseline level, but it remains less mature than the forward surrogate task and would likely benefit from stronger conditioning, more data, and more advanced generative models.

4.2.2 Generative Diffusion for Fab-Friendly Design (Qinlin)

The experimental phase of this research was designed systematically to answer two fundamental questions. First, we aimed to empirically verify whether a probabilistic generative framework could resolve the spatial averaging failure inherent to deterministic inverse design. Second, we sought to isolate how progressive architectural constraints within the generative pipeline influence both the mathematical accuracy and the physical manufacturability of the predicted waveguide structures.

To ensure unbiased evaluation, all models were trained and tested on an identically curated dataset, with 10% of the samples strictly reserved as a hold-out test set to prevent data leakage. The conditioning input for all evaluations remained constant: an unseen target S_{11} reflection spectrum, the period number, and the target scattering loss. The models were tasked with reconstructing the corresponding 100-point 1D transverse width profile.

Rather than presenting a single optimized model, we structured the experiments as a progressive ablation study to track the exact impact of each engineering decision. The evaluation protocol was conducted in three stages:

1. **Baseline Comparison:** We compared the deterministic Multi-Layer Perceptron (MLP v1) directly against the initial unconstrained diffusion model (DDPM v1). This isolates the fundamental difference between MSE-driven regression and probabilistic sampling.
2. **Generative Variance Analysis:** Because generative models map a single condition to a probability distribution, evaluating a single output provides an incomplete picture. For every DDPM evaluation, we queried the model five times using the exact same optical condition but varying the initial Langevin noise vectors (seed0 through seed4). This allowed us to visualize the model's ability to explore the "one-to-many" physical solution space.
3. **Architectural Ablation:** We conducted side-by-side inferences using DDPM v1, v2, and v3 on identical test samples. This isolates the physical impact of explicit interventions, namely spatial padding, thermodynamic boundary clamping, and CFG.

The success of these experiments was evaluated using a dual criteria approach. Quantitatively, we tracked the unnormalized spatial MSE between the predicted profiles and the ground-truth geometries. Qualitatively, we conducted visual inspections of the 1D profiles. In physical nanophotonics, a low MSE is meaningless if the resulting boundary is jagged or topologically unstable, as such artifacts induce catastrophic light scattering. Therefore, the smoothness and continuity of the generated boundaries were heavily weighted in our analysis.

Deterministic vs. Generative Baselines:

The initial comparison between the deterministic and probabilistic approaches is presented in Figure 4.12 for Sample 563. The ground-truth 2D mask (top row) exhibits a discrete, abrupt step-transition in the transverse width. The MLP v1 model (middle row) yields a high spatial MSE of 0.7678. Visually, the MLP prediction roughly tracks the macroscopic spatial envelope but generates highly erratic, jagged boundaries along the entire propagation axis, completely failing to capture a distinct structural step. Conversely, the DDPM v1 prediction (bottom row) achieves a significantly lower MSE of 0.3431. The 1D profile produced by the diffusion model forms a distinct, continuous structural cavity, largely devoid of the high-frequency pixelated noise seen in the MLP, though it exhibits irregular wave-like fluctuations rather than a flat step.

Generative Variance and Boundary Artifacts (DDPM v1):

To test the multimodal capacity of the unconstrained diffusion model, Figure 4.13 illustrates the profiles generated by DDPM v1 across four different optical designs (0030, 0077, 0568, and 0224). The model successfully generates a diverse spread of spatial profiles for the exact same input condition, confirming its generative capability. However, severe geometric anomalies are objectively present across the samples. For example, in Design 0030 and Design 0077, several generated profiles (such as the cyan line for seed4) exhibit abrupt, near-vertical drops in the transverse width midway through the structure. Additionally, severe boundary kinking and spatial collapse are frequently visible at the extreme edges of the propagation axis.

Spatial Stabilization (DDPM v1 vs. DDPM v2):

Figure 4.14 presents an ablation study isolating the effect of the spatial padding and mathematical clamping introduced in DDPM v2. In the unconstrained DDPM v1 plot (top), the previously noted boundary collapses and sharp edge hooks are prominent. In the DDPM v2 plot (bottom), these specific terminal artifacts are entirely eradicated. The five sampled profiles in v2 maintain perfect topological continuity and transition out of the spatial domain smoothly. However, the generative variance remains exceptionally broad; the five seeds spread widely across the transverse axis

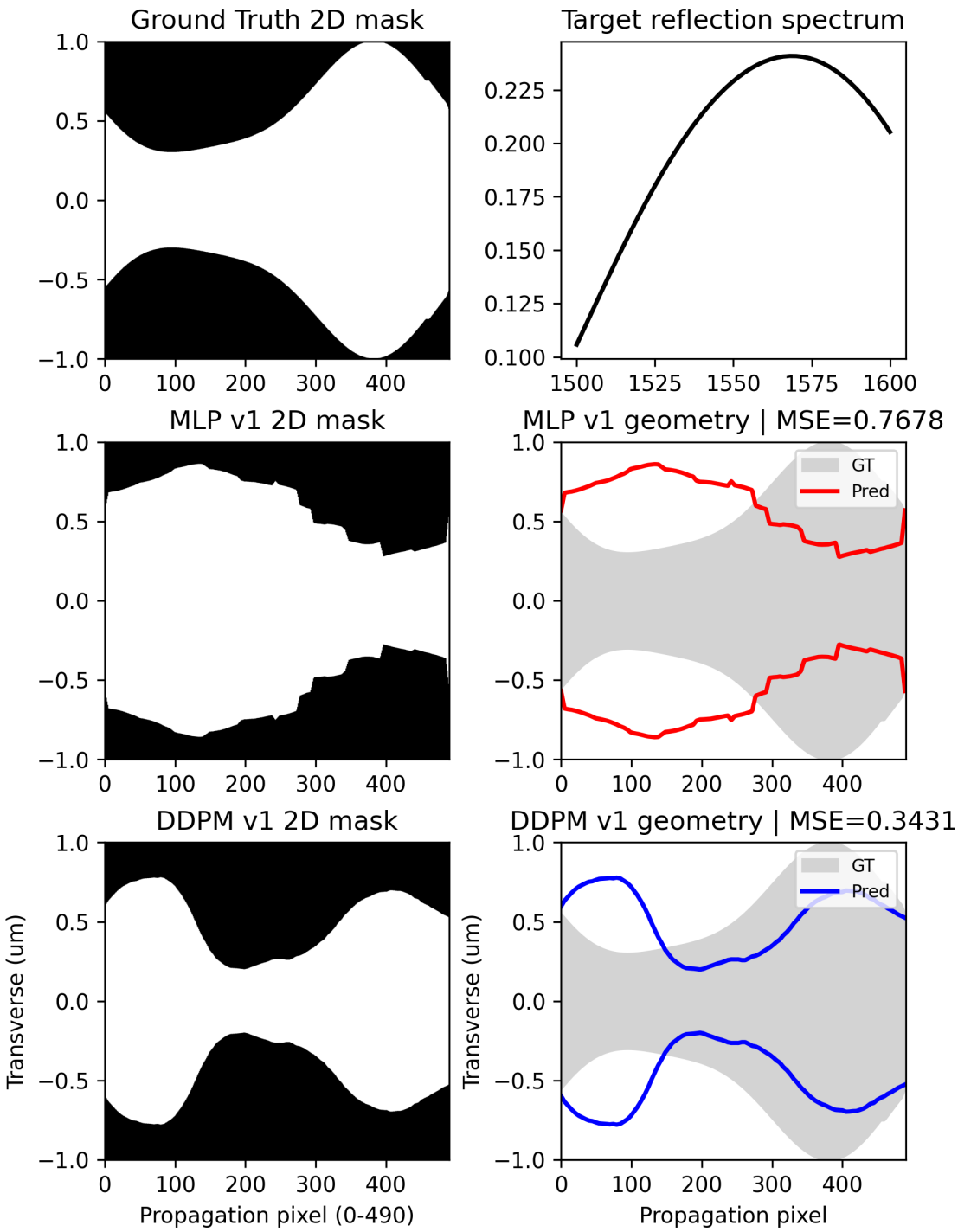

Figure 4.12: Comparison of MLP and DDPM models for Sample 563 (Period number: 10, Loss: 0.001). The top row displays the ground truth 2D mask and target reflection spectrum. The middle and bottom rows show the predicted 2D masks and geometry profiles for **MLP v1** (MSE: 0.7678) and **DDPM v1** (MSE: 0.3431), respectively. In the geometry plots, the solid lines represent the model predictions compared against the gray-shaded ground truth.

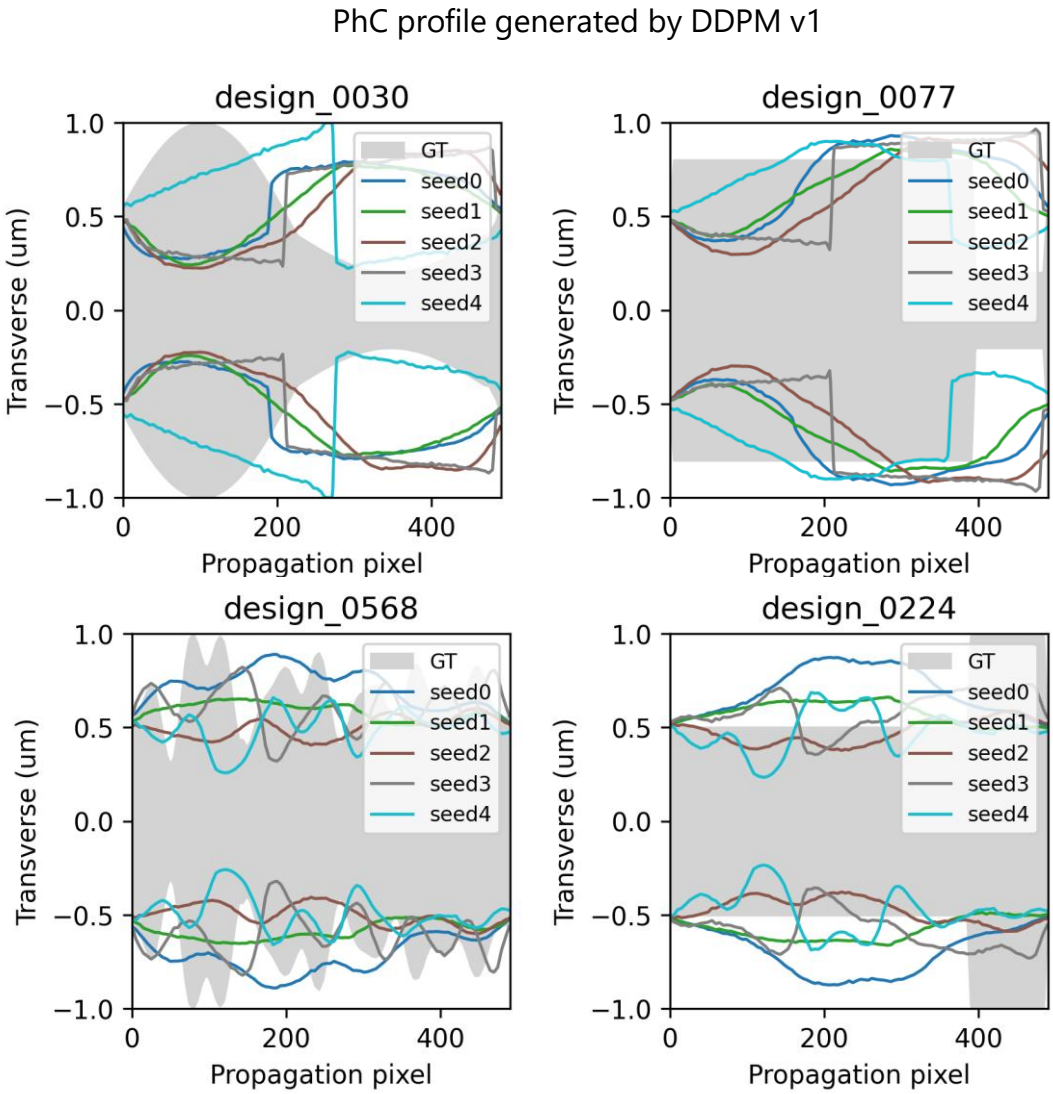

Figure 4.13: Photonic Crystal (PhC) profiles generated by the **DDPM v1** model for four different designs: **0030** , **0077** , **0568** , and **0224**. Each plot compares the ground truth (GT) geometry, shown as a gray-shaded region , against five separate profiles generated using different random seeds (seed0–seed4)

and generally fail to conform to the discrete, rectangular step-transition dictated by the gray ground-truth envelope.

High-Fidelity Guidance (DDPM v3):

The final architectural progression is captured in Figure 4.15, tracking the performance of v1, v2, and v3 across Designs 61 and 338. While v1 exhibits boundary artifacts and v2 exhibits high spatial variance, DDPM v3 introduces Global Self-Attention and CFG ($w = 3.0$). The objective result is a dramatic collapse in stochastic variance. Unlike the previous iterations, the five random seeds in v3 overlap almost perfectly. Furthermore, the generated profiles snap tightly to the ground-truth data, actively recreating the abrupt 90-degree spatial steps and significantly lowering the quantitative spatial MSE.

Discussion: The Paradox of Numerical Optimization

The progression of results from the MLP baseline to DDPM v3 provides clear answers to our primary research questions. However, a critical physical analysis of these generated geometries reveals a deep paradox regarding how machine learning models are evaluated in the physical sciences.

The highly irregular morphology of the MLP output in Figure 4.12 serves as empirical confirmation of the averaging effect in deterministic networks. Because inverse design is a one-to-many problem, an MSE-optimized MLP attempts to simultaneously satisfy conflicting geometric targets. The resulting compressed, noisy tube is the mathematical average of these configurations. In a physical cleanroom, a waveguide with such jagged sidewalls would suffer from massive out-of-plane scattering, rendering the device entirely non-functional. The ability of the diffusion model to sample distinct, continuous paths validates the necessity of generative architectures for multimodal physical problems.

Furthermore, the presence of mid-structure vertical cliffs and boundary hooking in DDPM v1 highlights a critical software-to-physics translation error. As hypothesized, the spatial indivisibility of the 100-point array caused compounding interpolation errors during the U-Net's reverse sampling phase. The ablation results prove that the explicit structural interventions in DDPM v2—specifically, spatial padding to 128 points and strict $[-1.0, 1.0]$ boundary clamping—successfully eradicated these artifacts.

However, the most profound insight emerges when we critically compare the highly optimized DDPM v3 against the intermediate DDPM v2. A naive interpretation of the metrics would suggest that DDPM v3 is the ultimate success of this research. It achieved the lowest MSE, collapsed the generative variance, and perfectly replicated the ground-truth step-geometry. Yet, a rigorous examination of physical engineering principles suggests this conclusion is flawed.

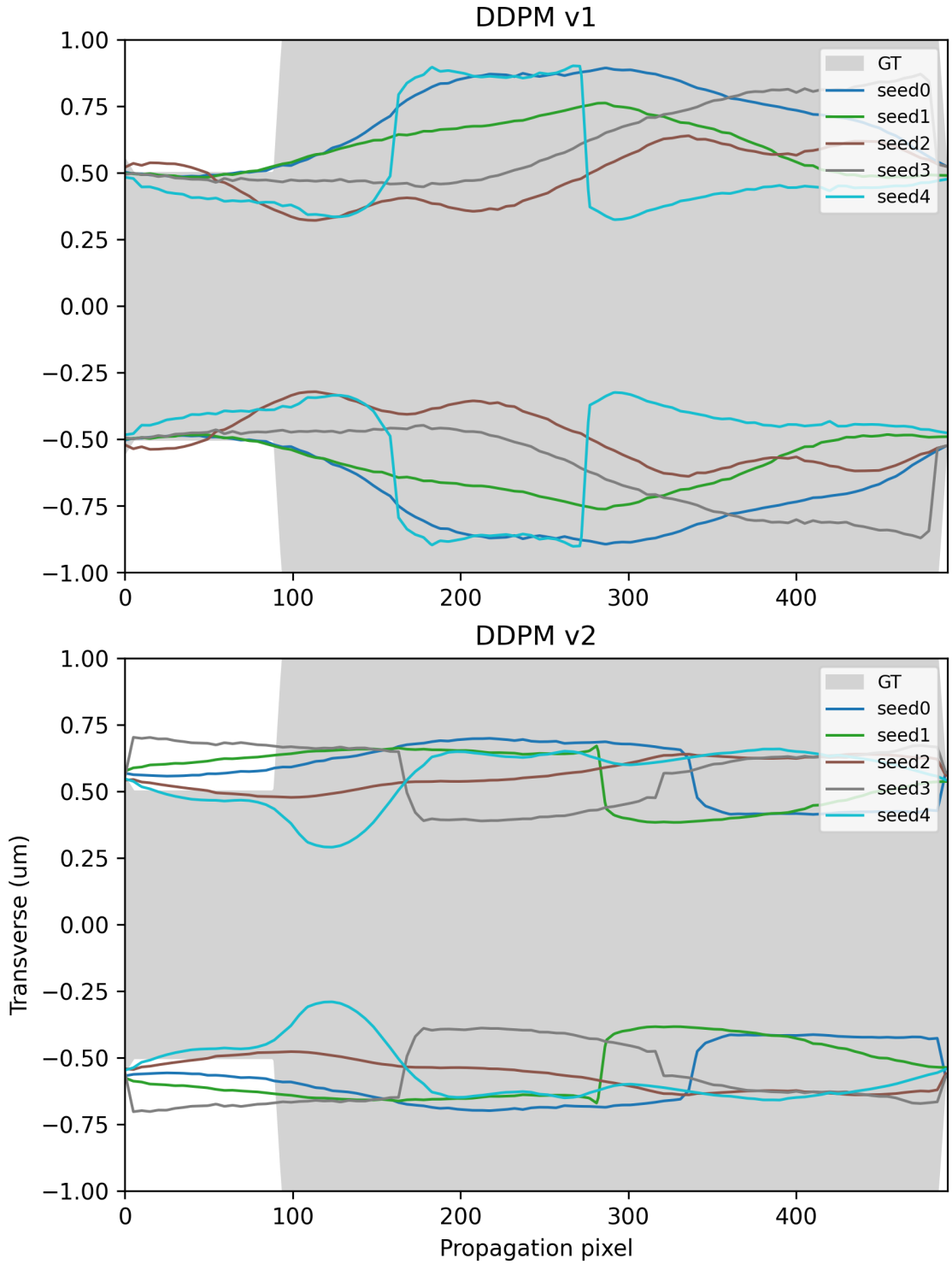

Figure 4.14: Comparison of stochastic profile generation between **DDPM v1** (top) and **DDPM v2** (bottom). The plots show the ground truth (GT) geometry as a gray-shaded region, overlaid with five unique profiles generated using different random seeds (seed0–seed4). The x-axis indicates the propagation pixel, and the y-axis shows the transverse dimension in micrometers.

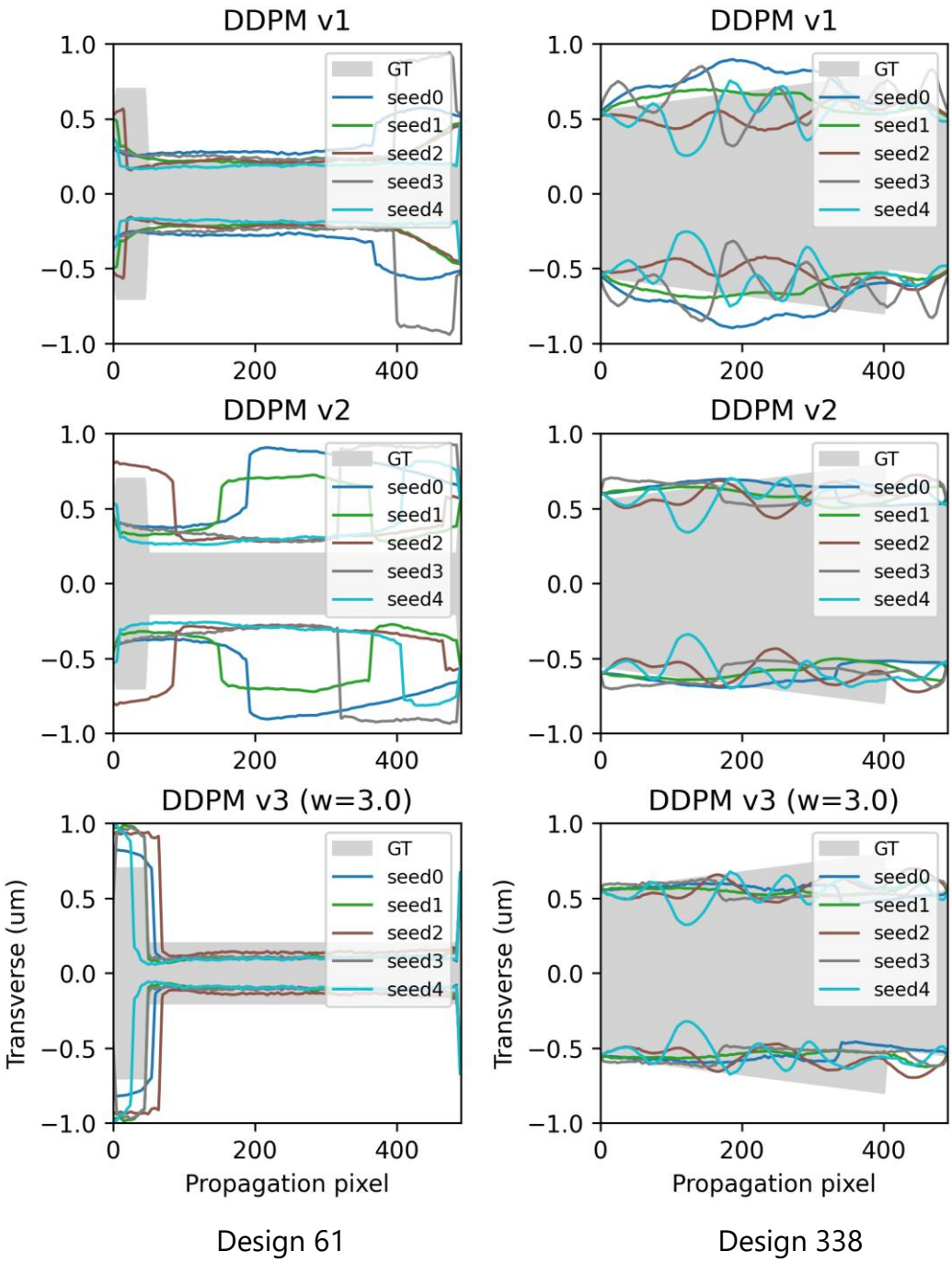

Figure 4.15: Comparison of profile generation across three model versions: **DDPM v1** (top row), **DDPM v2** (middle row), and **DDPM v3** ($w = 3.0$) (bottom row). The left column displays results for **Design 61**, and the right column shows **Design 338**. In each plot, the gray-shaded region represents the ground truth (GT) geometry, which is compared against five stochastic profiles generated using different random seeds (seed0–seed4)

The ground-truth datasets utilized in computational photonics are highly idealized. Abrupt, perfectly orthogonal 90-degree steps are prevalent because they align flawlessly with the Cartesian grids of Finite-Difference Time-Domain (FDTD) simulators, making bulk data generation computationally efficient. In a physical cleanroom, however, these shapes are disastrous. Deep ultraviolet lithography cannot produce perfect 90-degree internal corners due to natural exposure rounding, and abrupt waveguide discontinuities act as severe impedance mismatches that cause massive reflection losses.

By analyzing the output of DDPM v2, we observe a fascinating phenomenon: due to the natural spatial smoothing bias of its convolutional layers, v2 actively refused to draw the unphysical 90-degree steps. Instead, it leveraged its generative freedom to propose continuous, adiabatic tapers and smooth Bragg-like gratings. The diffusion model essentially "upgraded" the flawed training data into a much more fabrication-friendly layout, implicitly bridging the Simulation-to-Reality (Sim2Real) gap.

When we evolved the architecture to v3 and applied heavy guidance to aggressively minimize the MSE, we stripped the model of this generative freedom. Model v3 was mathematically forced to un-learn the smooth physics it discovered in v2, compelling it to memorize and reproduce the idealized, unmanufacturable artifacts of the FDTD dataset. Therefore, while DDPM v3 represents a triumph in algorithmic conditioning, DDPM v2 is arguably the superior model for actual physical fabrication. This exposes a critical misalignment in scientific machine learning: standard loss metrics penalize models for deviating from the dataset, even when the dataset itself is physically flawed.

4.2.3 Latent-Space Inverse Design (Milan)

The goal of these experiments is to evaluate whether the combined system, consisting of a geometry autoencoder, a forward surrogate model, and latent-space optimization, can generate geometries whose predicted spectral response matches a given target spectrum.

Experimental Setup

The geometry autoencoder was trained on the training split of the New-PhC dataset for 130 epochs. After training, both the autoencoder and the forward surrogate model were fixed and used in evaluation mode, meaning that their parameters were no longer updated.
For the inverse design experiments, ten samples from the test set were selected as target spectra y_{target}. For each target, the latent-space optimization was performed using 2000 optimization steps and $n_{\text{starts}} = 10$ independent initializations. The geometry x_{opt} with the lowest objective value across all runs was selected.

The predicted spectra $f_{\text{forward}}(x_{\text{opt}})$ were compared to the corresponding target spectra, and the resulting geometries were analyzed qualitatively.

Autoencoder Results

The autoencoder achieves stable reconstruction performance. The mean squared error decreases from approximately 1.02 (training) and 0.97 (validation) in the initial epochs to around 0.14 for both splits. The close agreement between training and validation losses indicates that the model generalizes well and does not exhibit significant overfitting.

This suggests that the autoencoder captures the dominant structural patterns of the dataset while maintaining a consistent representation across unseen samples. Although the reconstruction is not perfect, the achieved accuracy is sufficient to define a smooth and structured latent space.

This latent representation serves as a meaningful constraint for the inverse design process, enabling stable optimization while preserving physically plausible geometries.

Inverse Design Results

An ensemble approach with $n_{\text{starts}} = 10$ was applied for each target spectrum. The resulting objective values are typically in the range of 10^{-5} to 10^{-3}. For several targets, the optimization achieves very low errors below 10^{-4}, indicating near pointwise agreement between the predicted and the target spectra. Other targets show higher final errors in the range of 10^{-3}, corresponding to visible deviations in certain regions of the spectra.

Figure 4.16 shows representative examples of target and predicted spectra. In most cases, the global shape of the spectrum, including peak positions and general trends, is well reproduced. However, deviations are more pronounced in regions with stronger local gradients or sharper curvature.

The ensemble approach reveals that the optimisation process is highly sensitive to initialization (see Fig. 4.17). For a given target, the best loss values across different starts can vary by several orders of magnitude. Nevertheless, selecting the best-performing run consistently yields solutions with low spectral error across all targets.
The resulting spectral mean squared errors are significantly lower than the test error of the forward model, indicating that the optimization process can exploit the learned representation to closely match the target spectra.
Note that the loss values shown in the inverse design plots correspond to the spectral MSE, whereas the optimization itself minimizes a combined objective that includes

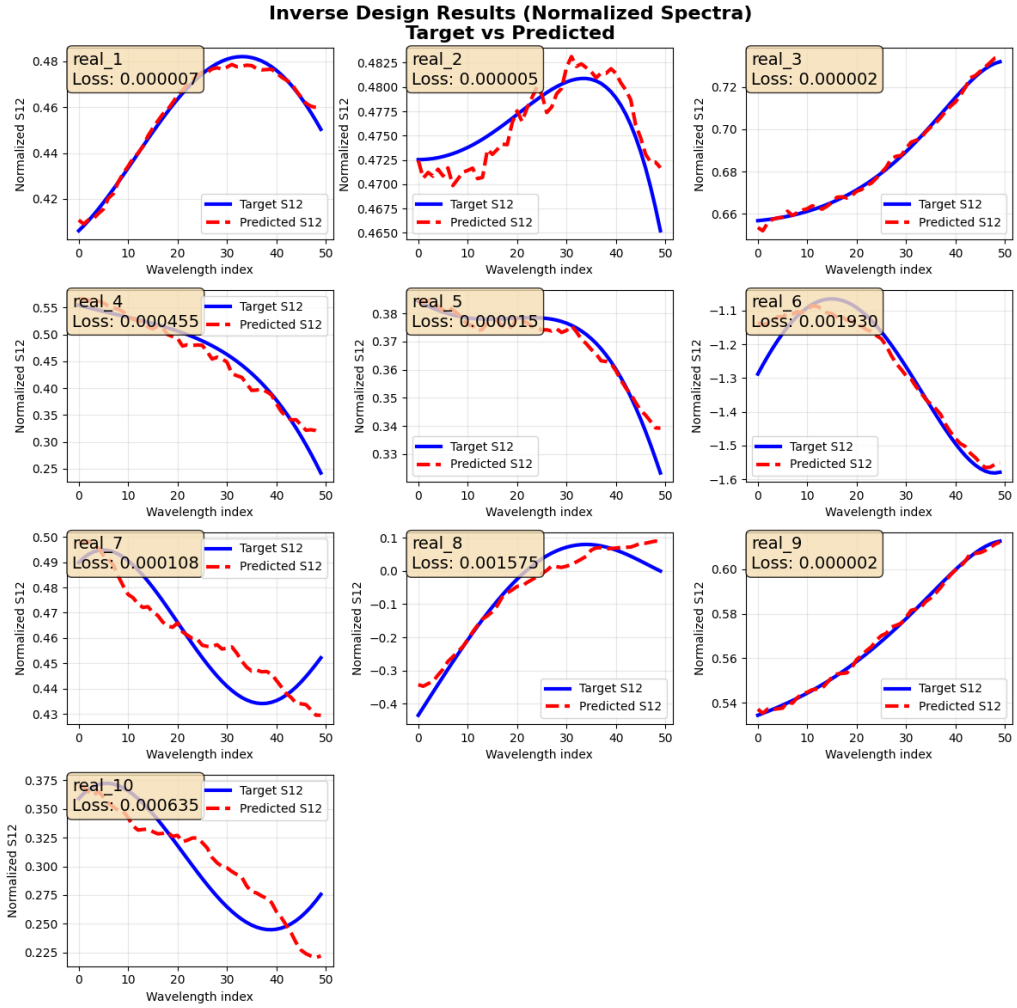

Figure 4.16: Target and predicted spectra with the corresponding spectral MSE. The reported loss reflects only the reconstruction error of the spectrum without regularization terms.

additional regularization terms.

Discussion

The results show that the combined system can reliably generate geometries that reproduce the global spectral characteristics of the target. In particular, for multiple targets, the target and predicted spectra closely match the target spectra. This suggests that the inverse design is effective in the learned latent space and that the relevant geometry–spectrum relationships are effectively captured.
At the same time, some examples show that the optimization is not equally successful for all targets. In these cases, visible deviations occur, particularly in regions with

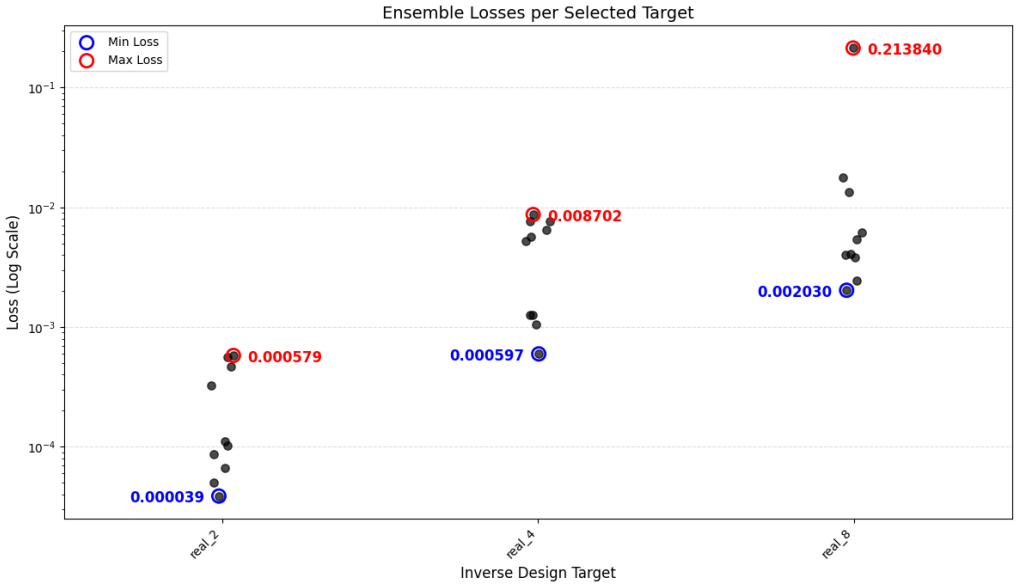

Figure 4.17: Final objective values across multiple initialization runs for selected inverse design targets (log scale). Each gray point represents one optimization run, while the blue and red markers indicate the minimum and maximum loss. The large spread of values highlights the sensitivity of the optimization process to initialization. The reported loss includes both spectral error and regularization terms.

stronger gradients or curvatures. This limitation directly propagates to the inverse design results, leading to larger deviations in regions with strong local gradients.
The use of the autoencoder is a key factor in the stability of the optimization. Restricting the search to the learned geometric space avoids non-physical or highly oscillating solutions. However, it should be noted that this restriction can also lead to limitations in solution accuracy, as not all possible geometries can be represented within this space.
The additional correction term Δx plays an important role here, as it allows the geometry to be refined locally, thereby achieving a better fit to the target spectrum. Without this term, the optimization would be more strongly restricted to the space specified by the autoencoder and could exhibit larger errors, particularly for more challenging targets.
The strong variability across different initializations reflects the non-convex nature of the optimization problem. Therefore, the ensemble approach using multiple initializations is crucial for reliably finding good solutions. By selecting the best run, the variability between different initializations can be effectively compensated for.
Overall, the results demonstrate that the proposed method is a robust and flexible approach to inverse design. Despite some residual deviations for individual targets, the method succeeds in reproducing the essential spectral properties, thereby providing a practical foundation for future design processes.

5 Conclusions and Outlook

5.1 Predictive Model (Ramtin)

This part of the project showed that the Predictive Model for Bandgap-Window Estimation is feasible. Instead of predicting the full band diagram as in Christensen et al.[2], the model was trained to predict compact bandgap-window quantities from the unit-cell image, namely the lower and upper edges of the first TM gap together with an auxiliary gap indicator. In this respect, the main objective of this predictive model was achieved: the CNN learned meaningful band-edge information directly from the 2D permittivity profile, and the regression results indicate that the simplified "filter-window" target can be learned with a relatively simple architecture. The upper edge was predicted more reliably than the lower edge, but both plots show that the model captures the correct overall trend.

At the same time, this part was only partially successful. While the regression component worked reasonably well, the binary gap/no-gap classification remained weak. The classifier achieved very high recall but very low specificity, meaning that it tended to predict a substantial gap too often. Therefore, the current model should be viewed as a proof-of-concept surrogate for estimating approximate bandgap edges, rather than as a reliable standalone detector of whether a substantial gap exists.

Overall, the results support the motivation of the Predictive Model for Bandgap-Window Estimation: predicting compact stopband descriptors is more practical and more manageable within the project scope than reconstructing the full dispersion relation. What was not achieved yet is the extension toward richer performance targets, such as direct stopband metrics in wavelength or the integration of the 3D simulation labels like R_{peak} and loss. These remain natural next steps for making the predictive model more directly relevant to realistic photonic-device design.

5.2 2D Image Tasks (Ramtin)

Overall, this project was able to answer the main research questions, although not equally strongly for all parts of the work. For the forward problem, the answer is clearly yes. The models were able to learn the relationship between the 2D refractive-index images, the period number, and the resulting spectral response with good accuracy. In particular, the reflection–transmission predictor achieved

low test error and reproduced the main spectral features reliably. The reflection–loss predictor also performed well for S_{11} and for average loss, although the full wavelength-dependent loss spectrum remained much harder to model. The main takeaway from this part of the project is that compact CNN-based surrogate models can provide a fast and stable approximation of expensive waveguide simulations.

For the inverse problem, the answer is more limited but still encouraging. The deterministic inverse model was able to generate geometries whose forward-predicted spectra remained reasonably close to the target behavior, which shows that it learned a meaningful inverse relationship. At the same time, the reconstructed images were often blurry and should not be viewed as exact or fabrication-ready solutions. The conditional variational autoencoder was a better match for the single-wavelength inverse setting, since it reflected the one-to-many nature of the problem by generating several different candidate geometries for the same optical target. This suggests that inverse photonic design is possible at a baseline level, but it is still much more challenging and less developed than the forward surrogate task.

Taken together, the results show that the strongest outcome of this project is the forward surrogate modeling rather than the inverse generation. The experiments demonstrate that machine learning can already be very useful as a fast predictive tool for photonic-crystal waveguides, especially when many candidate designs need to be evaluated efficiently. At the same time, they also make clear that inverse design under realistic constraints is more difficult, partly because the mapping from spectra to geometry is not unique and partly because some targets, such as the full scattering-loss spectrum, are noisy or incomplete. If this work were extended further, the most promising next steps would be to improve the amount and quality of training data, strengthen the conditioning of the inverse models, and explore more advanced generative architectures. In summary, the project leads to a strong conclusion for data-driven forward prediction and a more careful but still promising conclusion for inverse photonic design.

5.3 Fab-Friendly Design Conclusions (Qinlin)

This research was initiated to determine whether Denoising Diffusion Probabilistic Models could successfully navigate the multimodal landscape of inverse nanophotonic design, and how the architectural constraints of these models influence physical viability. Based on our systematic evaluations, we can answer these questions affirmatively, though with crucial methodological caveats.

We definitively conclude that deterministic regression models are structurally unsuited for inverse wave physics. The catastrophic spatial averaging demonstrated by the baseline MLP renders its predictions physically useless. By framing the design problem as a stochastic denoising process, the DDPM framework successfully decoupled geometry generation from strict one-to-one mapping, allowing the system

to propose diverse, continuous optical cavities.

Furthermore, we proved that generic computer vision architectures cannot be blindly applied to physical arrays. The transition from DDPM v1 to v2 demonstrated that explicit architectural regularizations—specifically symmetrical spatial padding for dimensional divisibility and thermodynamic boundary clamping—are strictly required to prevent the accumulation of numerical errors that manifest as physical scattering sites.

The most profound take-home message of this study is the identification of the metric misalignment paradox in physical machine learning. Our critical evaluation of DDPM v3 established that forcing a generative model to perfectly mimic an idealized simulation dataset often degrades the real-world manufacturability of the output. Convolutional diffusion models possess an inherent inductive bias that naturally smooths out the discrete artifacts of FDTD training data. Relying purely on traditional distance metrics like MSE fundamentally ignores the Sim2Real gap. In inverse physical design, topological continuity and lithographic feasibility must be prioritized alongside mathematical target fidelity.

5.4 Inverse Design via Latent-Space Optimization (Milan)

The proposed inverse design approach combines a learnt latent geometry representation with a differentiable forward surrogate model to enable efficient optimisation towards predefined spectral targets. The results demonstrate that this framework can reliably generate geometries that reproduce target spectra across a broad range of cases. In most cases, the predicted spectra closely match the desired responses, suggesting that the relationship between geometry and spectrum is effectively captured.

This directly addresses research question (iii), showing that data-driven inverse design methods can recover geometries for specified spectral responses. However, the results also reveal important limitations. Performance deteriorates for spectra with sharp local features or strong gradients, where noticeable discrepancies persist. These limitations primarily arise from the smoothness bias of the forward surrogate model and the restricted expressiveness of the learned latent geometry space.

The results also suggest that the method is highly effective at capturing global spectral characteristics, whereas fine-scale details are more challenging to reproduce. This highlights the fundamental trade-off between stability and expressiveness: while constraining the optimisation to a learned latent space improves robustness and physical plausibility, it can also limit the achievable accuracy for complex target spectra.

Overall, latent-space-based inverse design provides a stable and efficient framework within a constrained regime, but it is not robust across all target types. Future work should therefore focus on improving the forward model, for example by using larger architectures or physics-informed approaches, and on developing more expressive latent representations. Additionally, conditional generative models that map target spectra directly to geometries could offer a faster, more flexible alternative to iterative optimisation.

5.5 Outlook and Future Work

While the generative framework developed here successfully proposes continuous, boundary-stable geometries, the current evaluation methodology remains computationally isolated. To definitively prove the physical utility of these generated structures, the evaluation loop must be closed through empirical and optical validation.

The immediate next step required to validate this methodology is forward optical simulation. The design generated by the GNN must be imported back into an electromagnetic solver, such as Lumerical FDTD. Simulating the actual S-parameters of the AI-generated designs and comparing them directly against the original target spectra will objectively quantify whether the diffusion model successfully achieved the required optical resonance despite deviating from the square ground-truth shapes.

Following computational validation, the ultimate verification of this framework requires physical fabrication. Extracting the generated geometries as GDSII layout files, fabricating the photonic crystal mirrors via electron-beam lithography on Silicon Nitride wafers, and performing real-world optical characterization with a tunable laser will definitively bridge the gap between algorithmic generation and functional nanophotonic hardware.

Finally, the insights gained from this study dictate a fundamental overhaul of how future training data is curated. The generative models in this research were artificially constrained by training on FDTD-optimized datasets featuring discrete step-transitions. Future datasets must be generated using lithography-aware parameters. By constructing ground-truth geometries using continuous B-splines and simulating lithographic corner rounding during the data generation phase, we can provide the diffusion model with a physically accurate prior. Training the network on inherently fabrication-friendly data from the outset will eliminate the conflict between target fidelity and cleanroom manufacturability, fully unlocking the potential of conditional diffusion models for next-generation optical engineering.

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